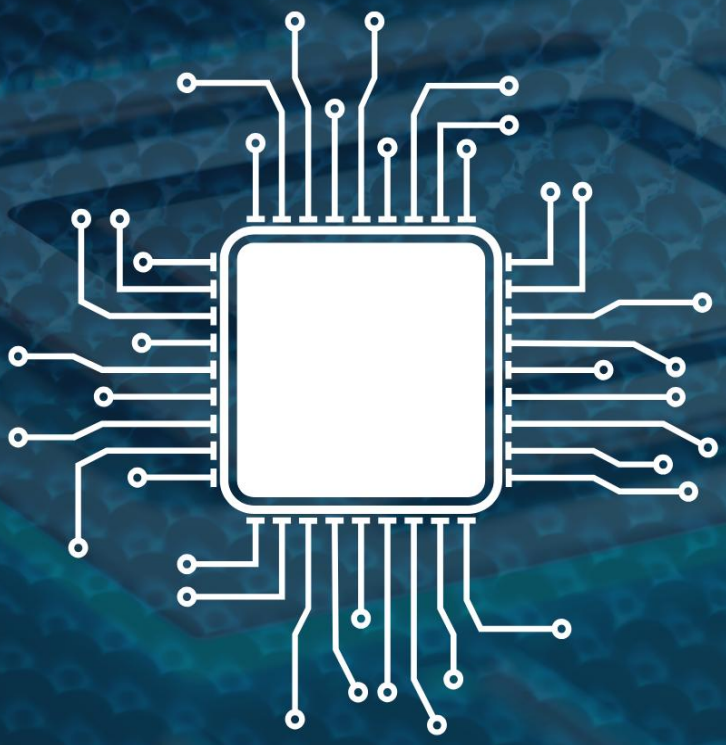
BASTION RESEARCH

SEMICONDUCTORS

INSIDE THE NANO WORLD



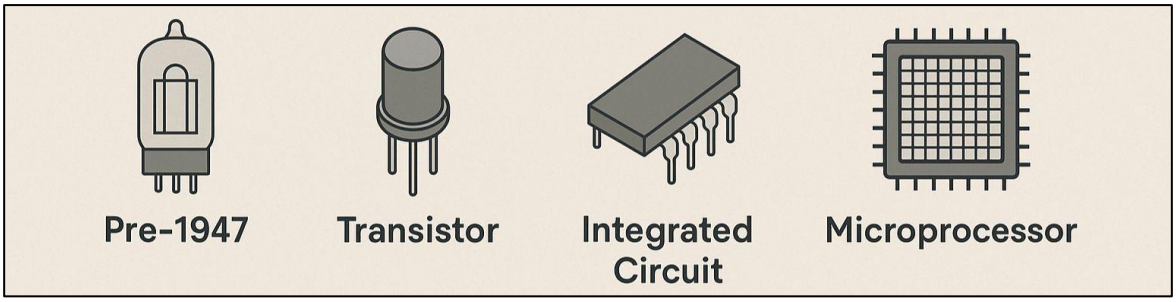
THE UNSEEN HEARTBEAT BEHIND TODAY'S DIGITAL WORLD

BASTION RESEARCH

Index | Semiconductor Series

1. History & Evolution of Semiconductors
2. Understanding The Semiconductor Value Chain
3. Key Players in The Semiconductor Value Chain
4. The Intel Story: From Silicon King to Underdog
5. TSMC: From a Radical Idea to a Global Semiconductor Titan
6. Advanced RISC Machines
7. Fabless, Fearless, and Worth Billions
8. India's Leap into the Semiconductor Future

History & Evolution of Semiconductors



The invisible engine

Semiconductors are the invisible engine behind modern life. You don't see them, but they're everywhere, in your phone, car, microwave, laptop, hospital scanner, power grid, and even in fighter jets. Think of them as the "brains" and "muscles" of electronics: they make things think (process data), remember (store data), and act (control machines). Comfortably, it's fair to say that Semiconductors are to the 21st century what steel was to the 19th and oil was to the 20th. They're the invisible infrastructure of our digital, connected, and automated world.

Out of sheer curiosity, we've been digging deep into the semiconductor space for a while. And over the next few editions, we'll share our insights with you, layer by layer, in plain English

Today, we begin at the very start: what electronic manufacturing looked like before semiconductors changed the game and who were the people involved in driving this change. Let's dive in…

Travelling back in time – The Age of Vacuum Tubes

Before the invention of semiconductors, electronics were big, slow, hot, expensive, and fragile. In the first half of the 20th century, the electronics industry relied on vacuum tubes, glass tubes that could amplify and switch electrical signals. Here's what electronics looked like back then:

Size: Devices were huge, room-sized computers and suitcase radios.

Heat & Power: Tubes ran hot and sucked power like crazy.

Reliability: They broke often and had short lifespans.

Cost: Expensive to build, assemble, and maintain.

Despite these limitations, vacuum tubes enabled early breakthroughs: The first radios and televisions, WWII radar systems, the ENIAC (1945), the world's first general-purpose computer, which used 17,000 tubes!

Source – Company, Bastion Research

History & Evolution | Semiconductor Series

Picture this: You needed a full table or even a whole cabinet to perform basic calculations. There were no transistors, so computers were rare, slow, and limited to governments, the military, and universities. There was no integration, everything was built using individual parts. Consumer electronics were luxury goods, not mass products.

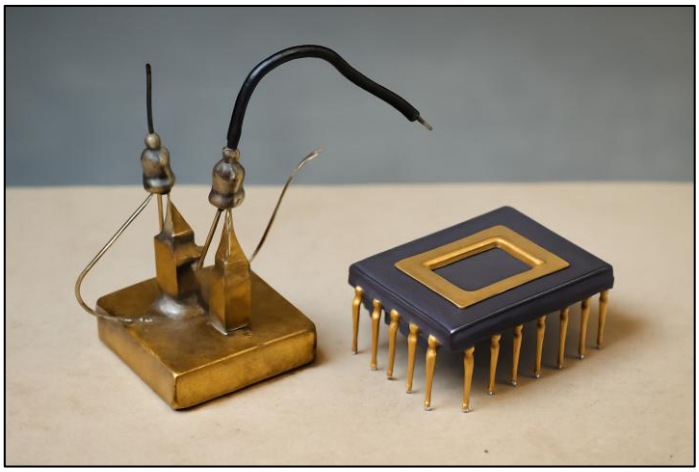
(Image of a vintage 1950s vacuum tube radio. It captures the exposed vacuum tubes, warm wooden cabinet, and the illuminated tuning dial, giving a clear representation of what early electronics powered by vacuum tubes looked like.)

Though vacuum tubes were a breakthrough in their time, it quickly became clear they weren't built for the future. The game changed when some of America's brightest minds came together at Bell Labs during World War II. Among their many inventions was something that would quietly reshape the modern world: the semiconductor. The sleek, scalable successor to vacuum tubes.

<u>The Breakthrough Invention – Birth of Semiconductors</u>

The idea that certain materials, known as semiconductors, could control the flow of electricity was proposed as early as the 19th century. Fast forward to 1947, John Bardeen, Walter Brattain, and William Shockley at Bell Labs invented the transistor. This breakthrough was the key moment in semiconductor history. The transistor could amplify electrical signals (like a vacuum tube) but in a tiny form, with lower power consumption, greater reliability, and no heat issues. The transistor was made from semiconductor materials, such as germanium and later silicon. By applying precise amounts of electrical current, transistors could control or switch electronic signals, doing what vacuum tubes did, but in a far more efficient and compact way.

Then came the next evolution, which truly unlocked the power of semiconductor materials and the transistor. In 1958-59, Jack Kilby at Texas Instruments and Robert Noyce at Fairchild Semiconductor are credited with inventing the integrated circuit (IC), which took semiconductor technology a significant step further. The invention of the IC allowed entire circuits (comprising many transistors) to be placed on a single chip, making computers, radios, and television sets smaller, cheaper, and more reliable. This was the birth of miniaturization.



Source – Company, Bastion Research

(Conceptual images of the first transistor (left) and the first IC (right) that transformed the electronics industry.)

While germanium was used in early transistors, silicon became the dominant material in semiconductor manufacturing by the 1960s. Silicon is more abundant, stable, and easier to work with compared to germanium.

Shockley Semiconductor and the Traitorous Eight

The story of Shockley Semiconductor and the Traitorous Eight is one of the most fascinating episodes in the history of the semiconductor industry. It revolves around a combination of innovation, internal conflict, and ambition that ultimately led to the creation of Fairchild Semiconductor, a company that played a central role in launching Silicon Valley and revolutionising the electronics industry.

William Shockley, a physicist and co-inventor of the transistor at Bell Labs (along with Bardeen and Brattain), became one of the pioneers of semiconductor technology. After winning the Nobel Prize in Physics in 1956 for his work on the transistor, Shockley left Bell Labs and set up his own company: Shockley Semiconductor Laboratory. The reason behind this move was that the transistor had proven to be a revolutionary technology, and Shockley was eager to develop and commercialise it further. This became the first company to focus on the commercial production of semiconductors.

His primary goal was to develop the technology that would lead to the creation of integrated circuits (ICs). To support him in this endeavour, he had a group of talented engineers and physicists who worked for Shockley Semiconductor. Shockley's management style was highly authoritarian, and he was known to be difficult to work with. He was very demanding and expected a great deal from his employees, which led to tensions within the company.

This resulted in a group of eight members leaving Shockley Semiconductor and starting their own company, Fairchild Semiconductor. This move resulted in Shockley giving them a derogatory name, the "Traitorous Eight."

History & Evolution | Semiconductor Series

Significance of Fairchild Semiconductor's role

The reason Fairchild Semiconductor was so important was that it **achieved a breakthrough in ICs**, which would go on to revolutionise electronics by allowing multiple transistors to be placed on a single chip. Fairchild became one of the first major semiconductor companies in the Palo Alto area, which would later come to be known as Silicon Valley. The company was a key player in the formation of Silicon Valley as a global hub of technology and innovation.

Additionally, two members (Robert Noyce and Gordon Moore) out of the eight went on to found INTEL, the company that powers most of PC that we use today. In short, Fairchild Semiconductor played a pivotal role in the mass production of integrated circuits (ICs) and the advancement of silicon technology, which paved the way for innovations ranging from personal computers to smartphones.

Key Differences

Feature	Transistor	Integrated Circuit (IC)
Composition	A **single component**	Contains **multiple transistors** and other components in one package
Function	Amplifies signals or acts as a switch	Performs more **complex tasks** (processing, signal conditioning, etc.)
Size	**Smaller** (but still larger than modern ICs)	**Tiny** and highly compact, as it integrates multiple components
Complexity	Performs a **basic** function (amplify/switch)	Can perform **complex operations** (like processing or memory storage)
Example	**Radio transistors**	**Microprocessors, memory chips, power ICs**

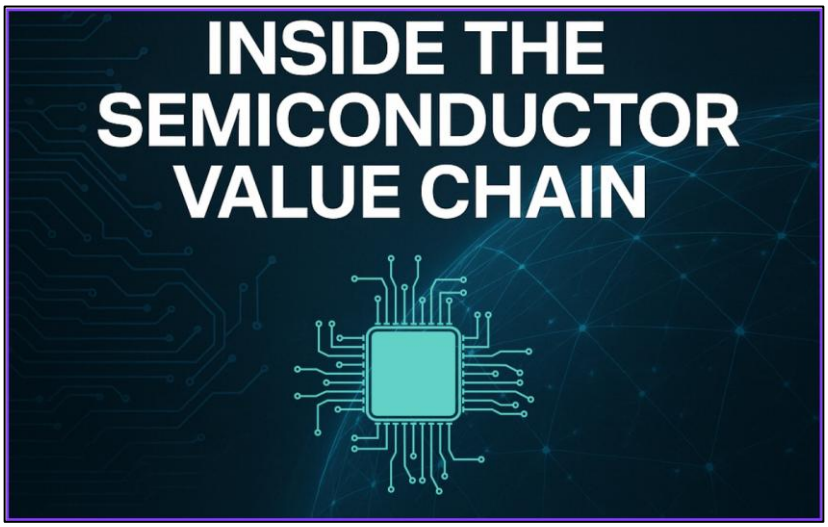
History & Evolution | Semiconductor Series

Then vs Now – Evolution of Semiconductors till Today

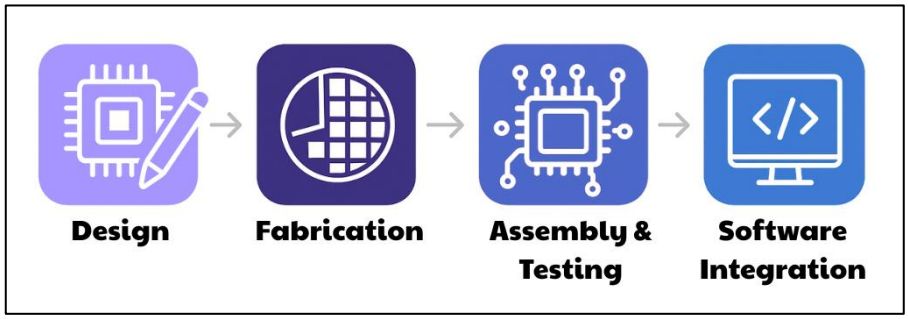
In 1971 (24 years after the transistor invention), Intel introduced the **Intel 4004**, the **world's first microprocessor**. This development allowed computers to be smaller and more affordable. By the **1990s**, semiconductor companies were able to pack **millions of transistors onto a single chip**, creating multi-core processors capable of performing thousands of calculations per second. Chips became the brain of everything from home entertainment systems to automated factories. Later in the 2000s, the introduction of smartphones was a pivotal moment. These chips integrated a **CPU**, a **GPU**, **radio-frequency chips**, and **memory** onto a **single device**, making **mobile computing powerful** and **portable**.

Today, semiconductors power everything from smartphones to cars, AI systems, space exploration, and even medical devices. Modern chips have **billions of transistors packed onto a single chip, with sizes shrinking to the nanoscale**. Modern chips are **tiny**, with semiconductor manufacturing now working at the **5nm** node. A **5nm process** means that the transistors on the chip are **5 billionths of a meter** in size, making them faster, more power-efficient, and more compact than ever. A single high-end smartphone today has a multi-core processor with more processing power than the supercomputers of the 1980s. Today, Semiconductors are truly the invisible engine of the electronic world.

Understanding The Semiconductor Value Chain



The Semiconductor Value Chain - FOUR Key Steps



1. Semiconductor Design

Designing a semiconductor is very much similar to building a house. Imagine you want to build a house. Before you lay a single brick, you hire an architect to design it. They decide things like, How many rooms you need, Where the doors and windows go, How the plumbing and electricity will run, etc… etc.

Chip Architecture: Before starting to design the chip, engineers need to answer a crucial question: Which kind of “language” or “Instruction” style will this chip understand? In technical terms, this is call the “Chip Architecture”. Think of it like rules for the computer brain, it tells a chip to read and execute instructions like, add these numbers or move this data or run this app, etc. Even more simply, chip architectures are different ways to talk to the processor. Without getting into complex details, following are TWO (Complex Instruction Set Computer - CISC and Reduced Instruction Set Computer - RISC or RISC-V) of the most commonly used chip architectures and some basic differentiating details:

Semiconductor Value Chain | Semiconductor Series

Summary:

Architecture	Style	Used In	Key Benefit
CISC	Complex	Intel/AMD PCs	Powerful instructions
RISC / RISC-V	Simple & fast	Smartphones, IoT	Efficiency, flexibility

While designing a chip, engineers design tiny electronic brains that live inside your phone, car, or laptop. Just like an architect, they use powerful softwares to plan things like, What the chip should do (e.g., process photos, connect to Wi-Fi, play games, etc), How the billions of tiny components (called transistors) are arranged on it, How fast and energy-efficient it will be, etc… etc… Practically, this process isn't as straightforward as it sounds. It takes months and sometimes even years to design chips.

Final Output: *A detailed file, like a construction blueprint which will be sent to a factory to begin making the chip (Fabrication).*

2. Fabrication

Once the architecture is finalised based on which the chip is designed, now comes the actual construction. The actual manufacturing of a chip happens in a ultra-clean, high-tech factory which in technical terms is called a "foundry".

Like we use bricks and cement to construct a house, foundrys use "Silicon Wafers". Simply explained, silicon wafers are thin, shiny discs made from super-pure sand. These wafers are processed using extremely precise machines in the presence of certain chemicals and gases.

Semiconductor Series

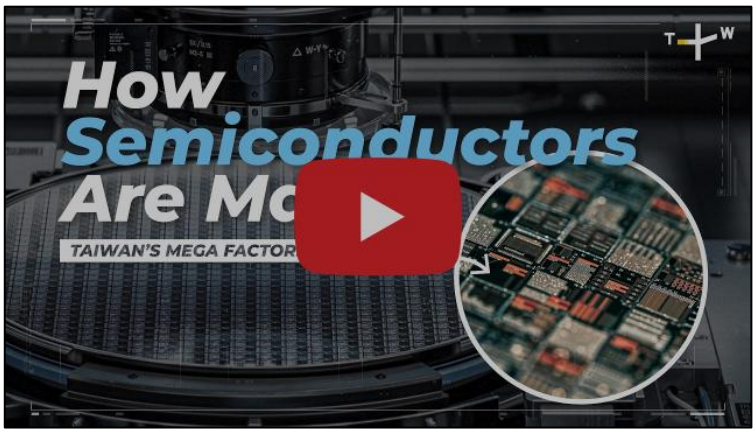
Summary:

Architecture	Style	Used In	Key Benefit
CISC	Complex	Intel/AMD PCs	Powerful instructions
RISC / RISC-V	Simple & fast	Smartphones, IoT	Efficiency, flexibility

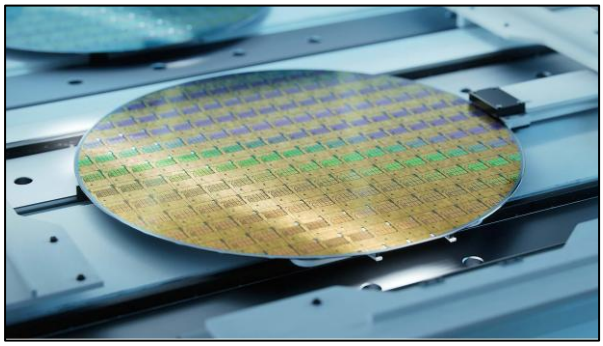
The blueprint (design) **is projected onto the wafer using light**, like stenciling. In this process, layers of metal and silcon compounds are added, patterned, and etched away. This process is repeated **hundreds of times**, adding millions or billions of transistors onto each chip. It's **like baking a 100-layer cake**, where you carefully stack ultra-thin layers of materials and carve out circuit patterns. What's fascinating is that each layer is 1,000 times thinner than a human hair.

Photolithography: It is a technical term for a tool that is **used to draw patterns**. In simple terms, **"Photolithography"** is using **light to draw incredibly tiny shapes** on the silicon wafer, shapes that eventually become transistors, wires, and logic gates inside the chip. Following are the steps involved in Photolithography.

Start with a clean wafer > Coating the wafer with photoresist > Shine extreme UV light through a mask > Light hardens parts of the photoresist > Wash away soft parts > Etch the pattern to form trenches or channels.



Final Output: *A silicon wafer full of many tiny, unfinished chips (called DIES).*



3. Assembly, Packaging & Testing - Making Chips Usable

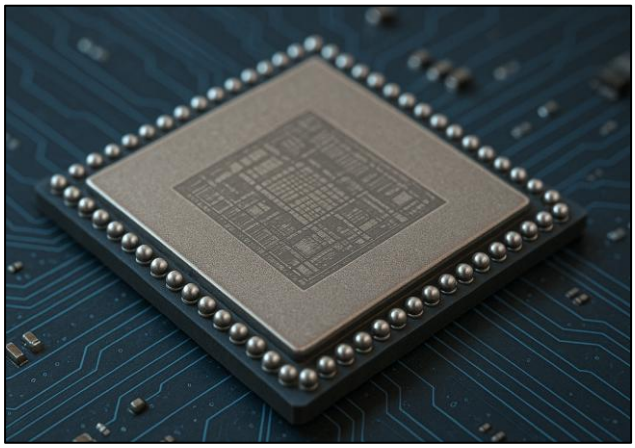
After the silicon wafer comes out of the foundry, it now has a **lot of miniature brains** on a wafer. But you can't plug a wafer into your phone, you need to slice, wrap, and test them before starting to use them. Following are the steps:

Cutting: The wafer is cut into individual dies (each one is a chip), just like cutting the cake from a tray into small pieces which can be sold individually.

Packaging: Each die is enclosed in a tiny plastic or ceramic case with metal legs or contact points. This protects the chip and allows it to be connected to a circuit board. To continue the cake analogy, each piece of the cake which was cut in the above step, is put into a wrapper.

Testing: Each chip is tested to check if it works and the broken ones are thrown away. Again the cake analogy, you check if any piece is burnt before shipping the same to customers.

Final Output: ***Finished, tested chips that can be sold and shipped***.



Semiconductor Value Chain | Semiconductor Series

4. System/Software Integration - Bringing Chips to Life

At this step, the chip is ready, but does nothing on its own. It's like a brain in a jar, powerful, but useless until it's put into a body and connected to nerves. Therefore, the chip is now placed inside a device (phone, car, laptop, etc.) and a software is written to control it. This is like telling the chip when to wake up, how to communicate with other parts, and what actions to perform.

For example, a chip in your phone that helps run your camera. The phone company writes a code that tells it how to adjust focus, optimize lighting, compress video, etc. In cars, chips manage everything from braking systems to music players.

Final Output: ***A complete, functioning device, powered by the chip.***

Conclusion

As we close out Part II of our exploration into semiconductors, it becomes clear that the journey from design to system integration is nothing short of an intricate, symphony of precision, and collaboration. Each step in the value chain, Design, Fabrication, Assembly & Testing, and System Integration, plays a vital role in powering the devices that define modern life. While Part I traced the roots of this industry and honored the visionaries who laid its foundation, this edition simply breakes down the value chain of the semiconductor ecosystem.

Key Players in The Semiconductor Value Chain

In the journey of exploring key players of the value chain, let's start with the FIRST step, SEMICONDUCTOR DESIGN:

As we explored in our previous chapters, chip design is one of the most complex and specialized aspects of creating the technology we rely on daily. When it comes to the players in this space, there are three key types to understand: 1) Chip Design Software Providers, who develop the tools that enable design; 2.1) Specialized Chip Architecture Designing & Licensing Companies, which focus on creating and licensing the blueprints for chips; and 2.2) Fabless Companies, which design chips but rely on third-party manufacturers to produce them.

1. Chip Design Software Providers: The Tools Behind Chip Design

Engineers and Designers use design software, also known as EDA (Electronic Design Automation) tools, to virtually test and create chips. These tools are essential in the chip design process, helping designers build complex chip blueprints and simulate their behavior in real-time.

These tools are the backbone of modern chip development. Without them, chip designers would be stuck trying to create these intricate designs manually, which would be both time-consuming and prone to mistakes.

Key Players - Synopsys (Market Cap ~$73 Bn), Cadence (Market Cap ~$81 Bn), Siemens EDA (formerly Mentor Graphics, Siemens' total market cap ~$179 Bn). The combined market capitalization of these top EDA companies is ~$333 Bn.

Next, we move to the companies that use these tools to actually design chips. At this stage, companies can be divided in TWO types: 2.1) Specialized Chip Architecture Designing & Licensing Companies and 2.2) Fabless Companies.

Key Players in the Value Chain | Semiconductor Series

2.1. Specialized Chip Architecture Designing & Licensing Companies

These companies design the core architecture, the blueprint of a chip and then license these designs to other companies. This makes it an industry standard for many devices we use today.

Why It's Special: These companies don't need to build factories or handle production costs. Instead, they earn in the form of royalties by licensing their designs to others, such as Apple and Qualcomm.

The Prime Example: A company called ARM is a prime example of such a company. In fact, ARM's designs power everything from smartphones to laptops, servers,and automobiles, making it the industry standard. If you're using a smartphone, laptop, or high tech car, chances are that ARM must have played a role in designed its chip.

Market Value: ARM's market cap is ~$153 Bn, but its influence is enormous. ARM's licensing model is highly profitable and powers billions of devices worldwide.

2.2. Fabless Companies

Meaning: Fabless companies are businesses that design semiconductors (chips) but don't actually make them. Instead, they outsource the manufacturing process to specialized companies called foundries (which we'll discuss next). These companies focus on the creativity and design.

They have mastered designing chips for various products, from smartphones to gaming consoles and AI systems. Their ability to focus on innovation while outsourcing production gives them the flexibility to lead in their respective fields and create new markets.

Key Players: NVIDIA (Market Cap ~$3.55 Tn), Qualcomm (Market Cap ~$169 Bn), AMD (Market Cap ~$206 Bn), Apple (Market Cap ~$3 Tn). The combined market cap of just these four fabless companies ~$7 Tn. Majority of the market cap belongs to Nvidia and Apple, the two largest corporations in the world.

The Interconnection of Chip Designers

There are inherent intricacies between these two types of designers because they are deeply connected. For example, ARM creates the fundamental blueprint (architecture) for chips, which it then licenses to fabless companies like Apple and Qualcomm. These fabless companies then tweak ARM's design as per their specific needs, adding proprietary elements to create the final chip design. This finalized design is then sent to foundries for manufacturing.

Now, let's move on to the next stage, **SEMICONDUCTOR FABRICATION**:

Key Players in the Value Chain | Semiconductor Series

To put the importance of this step into context, fabrication is the heart of chip production. Each chip design that is made is eventually sent to foundries for production. Companies that are involved in this step can be divided in TWO categories, 1) Focused Foundry Operators and 2) Integrated Players. Apart from them, it is also important to know about players who supply hi-tech machines, specialized chemicals and other raw materials to manufacturers which enable them to produce power-packed chips that empower our devices today.

1. Focused Foundry Operators

Modus Operandi: Many giant companies that design chips, like NVIDIA, Qualcomm and Apple, outsource manufacturing to Focused Foundry Operators. These companies specialize in the manufacturing of semiconductors which are designed by fabless companies (like Qualcomm or Nvidia). Think of it like a bakery that doesn't create the recipes (the designs) but bakes the cakes according to the recipes given by others. They operate large, advanced factories with cutting-edge technology to produce chips in large quantities for many different customers. They are experts in the manufacturing process, ensuring high quality and efficiency.

The reason why fables companies heavily depend on foundries for chip manufacturing is due to the following extreme realities:

Massive CAPEX: Building and maintaining a chip fabrication plant ("fab") is extraordinarily costly, costs often run in billions of dollars. The specialized machines required are equally expensive and demand highly specialized skillset to operate Foundries invest tens of billions in cutting-edge technologies such as EUV lithography, enabling the production of smaller, more powerful chips (e.g.,7nm, 5nm, 3nm), making them the preferred choice for manufacturing hi-tech advanced chips.

Key Players: Taiwan Semiconductor Manufacturing Company (TSMC), controls 60%+ of global semiconductor manufacturing and Global Foundries.

Unmatched Scale: Foundries like TSMC produce chips at immense volumes, a scale individual fabless companies cannot afford to replicate. Their unmatched production capacity allows them to serve hundreds of customers and churn out billions of chips annually, meeting the massive global demand for semiconductors. This enormous production scale allows foundries to achieve significantly lower costs per unit, making their operations extremely efficient and cost-effective.

Deep Expertise: Chip manufacturing demands a highly specialized workforce with profound expertise in material science, nanotechnology, and electrical engineering. Foundries like TSMC have accumulated decades of experience, developing unparalleled know-how crucial for accurately and reliably manufacturing complex chip designs with billions of transistors. Their continuous innovation in areas like smaller fabrication nodes (e.g., 3nm, 5nm) and advanced packaging is incredibly challenging to replicate.

These technical, economical, and manufacturing capabilities allow Focused Foundries to produce the most high-tech chips at the scale customers require, and at the most effective cost possible. So, even if Apple or NVIDIA wanted to manufacture their own chips, they would most likely be unable to produce such high-tech chips at the required scale and cost.

2. Integrated Players

Despite the above mentioned complexities of the semiconductor industry, a few standout companies have successfully managed to integrate both SEMICONDUCTOR DESIGN and MANUFACTURING in-house and achieve immense success.

Key Players: Intel and Samsung Electronics

Intel designs its own chips (like Intel processors) and also operates its own foundries to manufacture these chips in-house. This makes Intel an Integrated Device Manufacturer (IDM), as it controls both the design and manufacturing processes. Samsung also follows a similar model. Samsung designs its own chips (like Exynos processors) and also operates foundries that produce chips, including those for other companies. Therefore, Samsung is also an IDM. In summary, Intel and Samsung are integrated players because they handle both chip design and manufacturing, unlike fabless companies that only design chips and outsource manufacturing to foundries.

Market Value: The combined market value of three foundries, TSMC, Samsung, and GlobalFoundries, is ~$1.2 Tn.

Before we move to the next step, let's take a detour and talk about the machine and chemical providers to these foundries. This part of the value chain is also as concentrated as the foundry operation.

Machine Providers to Foundries

The machines used in semiconductor manufacturing, like photolithography, etching, and inspection machines, turn raw materials, such as silicon wafers, into functional chips. ASML (Netherlands) leads the EUV lithography market (with almost 100% market share), enabling the production of advanced chips at tiny sizes like 3nm. Additionally, Applied Materials and Lam Research provide essential tools for chemical deposition, etching, and cleaning, ensuring the chips are made with high precision. Their technologies are hard to replicate due to their complexity, high costs, and exclusive patents.

Key Players in the Value Chain | Semiconductor Series

Chemical and Material Providers to Foundries

Foundries use specialized chemicals and materials to clean wafers, layer circuitry, and deposit gases. Companies like **Merck, Air Products,** and **Dow Chemicals** provide key materials, such as **silicon wafers, gases**, and **cleaning chemicals**, essential for making precise chips. The complexity of producing these materials, along with their strong global supply chains and years of research, makes it very hard for new companies to match their expertise.

Key Players - ASML (Market Cap ~$294 Bn), Applied Materials (Market Cap ~$136 Bn), Lam Research (Market Cap ~$116 Bn), Merck Group (Market Cap ~$55 Bn), Air Product (Market Cap ~$60 Bn), Dow Chemicals (Market Cap ~$20 Bn). The combined market value of machine and material provider to foundries is $681 Bn and are integral to the semiconductor industry.

Now, moving to step-three, **ASSEMBLY, PACKAGING, & TESTING**

A manufactured chip, initially, is just silicon which needs further steps to make it usable. It's first assembled onto a base (substrate), then packaged in a protective cover, and rigorously tested for electrical performance and durability. These stages are vital for its correct function in any device.

While these steps may seem less glamorous than designing or manufacturing chips, they are critical for ensuringthat chips work properly and can be safely used in devices like smartphones,computers, and cars.

Key Players: ASE Technology (Market Cap ~$20 Bn), Amkor Technology (Market Cap ~$5 Bn), JCET Group (Market Cap ~$8 Bn), Together, the combined market cap of these companies is around $33 Bn.

Now comes the last part of the value chain: **SYSTEM INTEGRATION**, bringing chips to life

After semiconductor chips are manufactured, they still require one final step (System/Software Integration) to become usable in real-world devices.

Once integrated, the chip connects to essential device components like the screen, battery, memory, etc. Softwares, including the operating system and drivers, are then optimized to work seamlessly with the chip for efficient task performance. Finally, the entire device is thoroughly tested to ensure all components and software function together are running smoothly.

These end function is carried out by none other than OEMs like Apple, Samsung, Microsoft, Dell, etc. These companies take the final chip products and integrate them into their devices, ensuring the chip works flawlessly with other components and the device's operating software.

Conclusion

The semiconductor industry is a strange beast. Zoom into any one step, fabrication, chip design, packaging, and it feels like a monopoly. Just a handful of giants dominate, holding deep expertise, decades of IP, and near-insurmountable moats.

But zoom out, and it's anything but lonely. The full chip-making process is a global relay race of hundreds of companies across dozens of countries passing the baton, each doing one ultra-specialised job to bring a chip to life.

It's this bizarre mix of tight concentration and broad collaboration that makes semiconductors both incredibly powerful and incredibly fragile.

The Intel Story: From Silicon King to Underdog

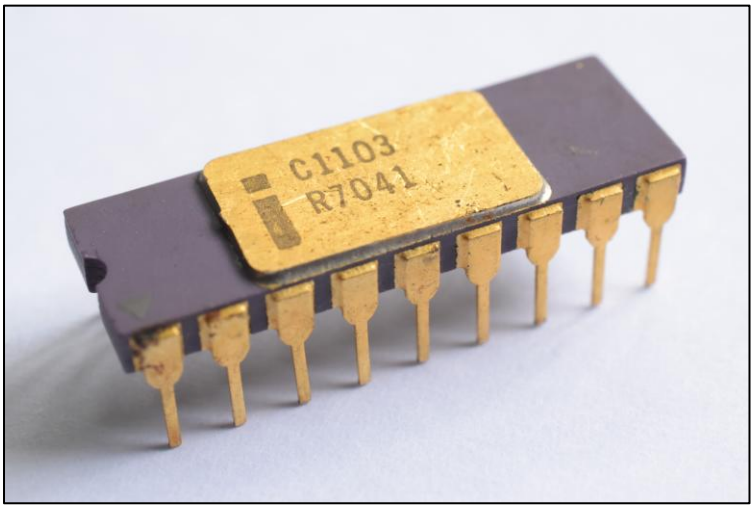
The Spark Behind the Silicon – The Birth of Intel

Intel was founded in 1968 by Robert Noyce and Gordon Moore, two of the legendary "Traitorous Eight" who helped build Fairchild Semiconductor. At Fairchild, they played a central role in inventing the planar integrated circuit — the foundation of all modern chips.

Frustrated by growing bureaucracy at Fairchild, they struck out on their own. Backed with $2.5 million from venture capitalist Arthur Rock, they started Integrated Electronics Corporation, soon shortened to Intel.

One of their earliest hires? Andy Grove, who would later become CEO and the cultural architect of Intel's high-performance ethos.

If you're assuming Intel jumped straight into microprocessors, hold that thought. In its early days, Intel was all about memory chips — both Static RAM (fast but expensive) and Dynamic RAM (cheap and scalable). In 1970, Intel launched the 1103, the world's first commercially available DRAM chip, which began replacing bulky magnetic core memory. By 1973, Intel had become the global leader in memory chips.



The Intel Story | Semiconductor Series

A Twist in the Tale – The Accidental Microprocessor Revolution

In 1971, a Japanese calculator company (Busicom) asked Intel to design 12 custom chips. Intel's engineers, led by Ted Hoff and Federico Faggin, proposed a radical idea: instead of 12 chips, build one general-purpose chip that could be programmed for different tasks.

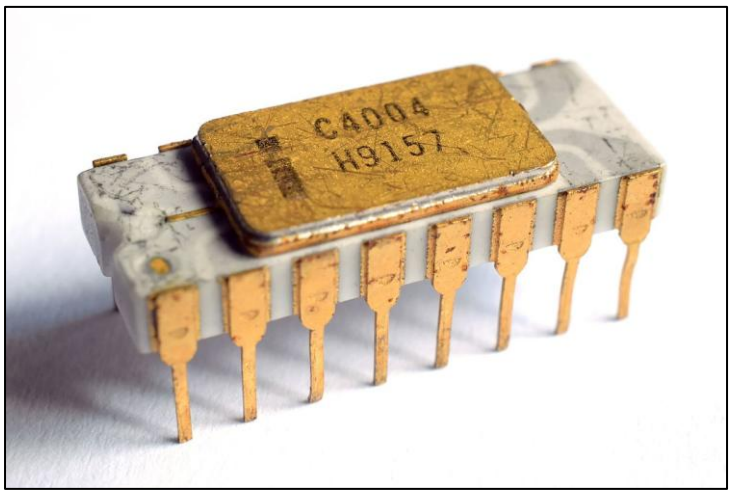
That became the Intel 4004, the world's first commercially available microprocessor.

At the time, it was just for calculators. But Intel saw the broader potential. Quietly, the 4004 planted the seeds of the personal computing revolution. That said, Intel's core business remained memory, at least for a few more years.

Quick Side Note #1: Confused between "chip" and "microprocessor"? Think of a microprocessor as one type of chip, just like memory, graphics, or sensor chips. All microprocessors are chips, but not all chips are microprocessors.

Quick Side Note #2: Each chip houses many, many transistors (think of them as tiny switches on a chip). With the advent of technology, chip sizes are getting smaller i.e. a 3nm (nm - nanometer) chip will have more transistors in the same area compared to a 5nm chip. Therefore, electricity moves faster, and the device uses less power and releases less heat compared to before.

Quick Side Note #2: Each chip houses many, many transistors (think of them as tiny switches on a chip). With the advent of technology, chip sizes are getting smaller i.e. a 3nm (nm - nanometer) chip will have more transistors in the same area compared to a 5nm chip. Therefore, electricity moves faster, and the device uses less power and releases less heat compared to before.



The Intel Story | Semiconductor Series

Intel Goes All In on Chips – Enter x86

Around 1975, Japanese firms began undercutting Intel on memory chips. In response, Intel pivoted hard from memory to microprocessors. In 1978, it launched the 8086, a 16-bit chip that laid the foundation for the x86 architecture still the backbone of most PCs today.

In 1981, IBM selected Intel's 8088 processor to power its first personal computer. The partnership exploded Intel's visibility. To encourage competition, IBM pushed Intel to license the chip to others (like AMD), but Intel's design leadership kept it ahead.

By the 1990s, Intel had become the heart of the PC world. It launched the Pentium in 1993, a branding masterstroke and its "Intel Inside" campaign turned a behind-the-scenes supplier into a household name. At its peak, Intel held over 80% of the desktop CPU market. It was, without question, the king of the computing hill.

Back then, performance was everything. Power efficiency? Not a concern as desktops were always plugged in.

Intel's Blind Spot – The Rise of Mobile

During the 2000s, Intel was thriving — but missed the quiet signals of disruption.

Intel's Blind Spot – The Rise of Mobile

During the 2000s, Intel was thriving — but missed the quiet signals of disruption.

At the time, mobile computing meant laptops, where Intel was strong. Mobile phones? A niche. Smartphones were clunky, lacked real internet access, and were still seen as luxury items. Intel doubled down on its core belief: Faster chips = Better PCs = Continued dominance.

But while Intel was cruising, the world was changing. Two undercurrents gathered momentum:

1. The arrival of faster mobile networks (3G in 2004–05)

2. The demand for smaller, power-efficient chips

2007–08: The iPhone and Android Era Begins

The iPhone wasn't the first smartphone, but it felt like the future, full touchscreen, slick interface, real internet, media, and later, apps. Android followed quickly, and suddenly the world didn't crave faster desktops; it wanted portable, efficient, and always-connected devices.

Performance took a back seat. Power efficiency became king.

These new devices needed chips that used less energy, ran cooler, and fit into tight spaces. That wasn't Intel's strength, but it was ARM's.

ARM chips weren't as powerful, but they were: a) Far more power-efficient, b) Highly customizable and, c) Perfect for mobile

Every major smartphone chip, Apple's A-series, Qualcomm Snapdragon, Samsung Exynos, was based on ARM.

A famous misstep? In 2006–07, Apple offered Intel a chance to design chips for the iPhone. Intel said no, the margins looked slim, and the volumes were uncertain. That chip later became the Apple A4, built on ARM and manufactured by Samsung. The rest, as they say, is history.

How Did Intel Lost Its Way?

Before getting to the above topic, let's revisit what we covered last week: Intel operates as an Integrated Device Manufacturer (IDM) — it both designs and manufactures its own chips.

1. Losing the Design Battle

Fearing ARM's rise, Intel tried to shrink its x86 chips to fit smartphones and tablets. It partnered with Lenovo, ASUS, Motorola and ran massive subsidies to push Atom processors. But Atom chips still consumed too much power and underperformed ARM products.

Even Apple didn't bite. Despite multiple efforts, Intel couldn't convince it to replace ARM, not even in entry-level iPhones or iPads. After nearly a decade of struggle, Intel shut down its smartphone processor business in 2016. Then-CEO Brian Krzanich candidly called it Intel's "greatest mistake."

2. Losing the Manufacturing Edge

Intel didn't just stumble on design, it also lost its long-held manufacturing lead. For years, Intel led the industry in process node innovation. From 90nm to 14nm, it was ahead of everyone. But its 10nm launch, planned for 2016, was plagued by yield issues (more defective chips than desired) and delays. As a result, it shipped in volume only around 2019–20.

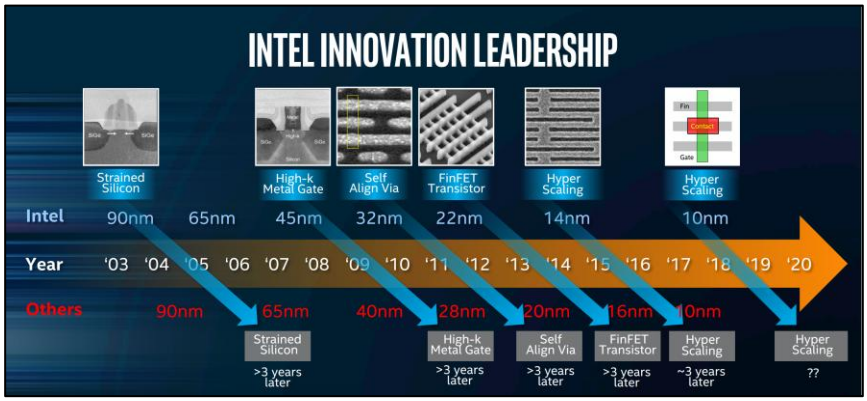
Meanwhile, TSMC pulled ahead by launching:

- 10nm in 2016

- 7nm in 2018

- 5nm in 2020

- 3nm by 2023



While Intel was stuck on 10nm, TSMC powered ahead. AMD, Apple, Nvidia, and others rode that wave. For example, AMD's Ryzen chips (on TSMC's 7nm) outperformed Intel. Apple's M1 chip, on 5nm, shocked the world with its efficiency. Major cloud players also followed suit, moving away from Intel chips and fabs.

This is how Intel lost its edge in both design (to ARM) and manufacturing (to TSMC).

Can Intel Make a Comeback?

After missing the smartphone boom and losing its manufacturing dominance, Intel is now attempting one of the most ambitious turnarounds in tech history.

Under CEO Pat Gelsinger, the strategy is bold and three-pronged:

1. Reboot chip design with modular architectures like Meteor Lake

2. Rebuild world-class fabs across the U.S. and Europe

3. Open Intel Foundry Services (IFS), making chips not just for itself, but also for others, including rivals

This new approach is called **IDM 2.0**. In Gelsinger's words, Intel wants to be both the chef and the kitchen, and rent the kitchen out too.- 3nm by 2023

The Intel Story | Semiconductor Series

Billions are being invested. Intel aims to reclaim the manufacturing lead by 2025. It's not just about powering laptops anymore, it's about enabling everything: data centers, AI factories, and beyond.

Will it work? Time will tell. But one thing's for sure, Intel isn't going down quietly.

TSMC: From a Radical Idea to a Global Semiconductor Titan

Now, let's go through the dynamic story of Morris Chang, the founder of TSMC and how this company on a small island of Taiwan became one of the most important company of the 21st century (this is not an exaggeration).

In 1987, a 56-year-old engineer who most of Silicon Valley had written off, quietly launched a company in Taiwan with no products, no legacy, and no blueprint to follow. That company, TSMC, didn't design chips, didn't compete with clients, and didn't look anything like the semiconductor giants of the day. And yet, by flipping the industry's business model on its head, it unlocked a wave of innovation that powered the rise of Apple, Nvidia, and every major fabless firm since. Today, TSMC manufactures over 90% of the world's most advanced chips, making it arguably the most important tech company you've never heard of. This is the story of how it got there.

The Beginning: It's Not Over Until It's Over

Born in war-torn China in 1931, Morris Chang spent his early years moving from city to city as conflict and political upheaval swept the country. At 18, he left for the United States, beginning at Harvard before transferring to MIT, drawn by a deeper interest in engineering. That curiosity took him to Stanford, and eventually to Texas Instruments (TI), where he would spend the next 25 years building the company's semiconductor manufacturing muscle.

- Yes, he wasn't just managing operation, he built them:
- Oversaw TI's early entry into mass production of silicon transistors.

- Played a key role in setting up TI's first semiconductor fab in Asia.

Pioneered the concept of cost-effective high-volume chip manufacturing, which would later become the backbone of TSMC's model.

But despite his achievements over 25 years, he was not selected as a CEO of TI. Chang was passed over, more than once. At age 52, after decades of loyalty, he left TI. A brilliant career, but not the fairytale ending most would expect.

Return to Asia: Founding of TSMC

In 1985, the government of Taiwan had a bold ambition: transform the island from a low-cost electronics assembly hub into a high-tech manufacturing powerhouse. But building a semiconductor industry from scratch required more than just political will, it needed leadership, credibility, and someone who had already walked the path.

So they reached out to Morris Chang, then a semi-retired executive teaching at Stanford, with no personal or professional ties to Taiwan. He was 54. Most would've seen it as a quiet end to a decorated corporate career. He saw it as the beginning of something bigger.

Taiwan's Industrial Technology Research Institute (ITRI) offered him the platform to create something of his own. Two years later, in 1987, Morris proposed a radical idea:

- A semiconductor company that would manufacture chips but never design them.
- A company that would serve any client, large or small.
- One that would never compete with the companies it served.

At the time, it was unheard of. All major chip companies like Intel, TI, Motorola, were vertically integrated. They designed chips, built fabs, and tightly guarded their IP. Outsourcing manufacturing was almost laughable.

But ideas alone don't build factories. Morris needed funding and access to cutting-edge semiconductor technology. The Taiwanese government agreed to fund 50% of the proposed venture. The rest? He had to find it himself.

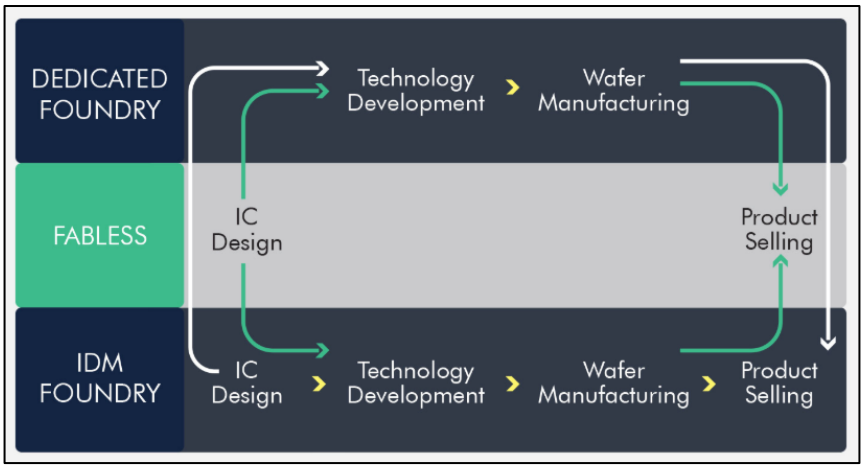
So he called on old contacts. One of them was Philips Electronics, yes, the same Philips you know from your home appliances. Philips not only invested but also contributed patents, equipment, and technical know-how, giving the project a crucial technological head start.

So with $220 Mn in initial capital, backed by a public-private partnership, borrowed IP, and a business model no one had dared to try, TSMC was born.

TSMC, The Titan | Semiconductor Series

The Execution Years: TSMC in the 1990s and 2000s

In 1987, TSMC introduced the first "pure-play" foundry, meaning it focused solely on manufacturing chips for other companies, rather than designing and selling its own products.



(Source: TSMC Website)

In the early 1990s, TSMC's edge wasn't innovation, it was execution. While companies like Intel and Texas Instruments raced to invent the future, TSMC focused on reliability, yield, and trust. That made it the go-to partner for a new generation of fabless chip companies, starting with Weitek, its first customer, and later LSI Logic.

At the helm was Morris Chang, still CEO, who made it clear: TSMC would succeed not by designing chips, but by becoming the most reliable and neutral manufacturer in the industry.

It didn't take long for that bet to pay off.

By 1994 itself, TSMC grew like wildfire became the first Taiwanese semiconductor company to list on the NYSE (as an ADR), cementing its global credibility. Over the rest of the decade, it mastered one process node after another: from 1µm to 0.6µm, then 0.35µm, 0.25µm, and 0.18µm, a pace that matched the best in the world.

Rather than trying to lead R&D, TSMC poured resources into manufacturing infrastructure: fully automated fabs, cleanroom upgrades, and statistical process control. These gave it precision on par with global IDMs, and the flexibility to handle a growing mix of customers and chip designs.

In the early 2000s, it made two key technical leaps.

0.13µm copper interconnects, which enabled faster, lower-power chips.

Preparing for FinFET transistor architectures, laying the groundwork for its future 16nm node.

TSMC, The Titan | Semiconductor Series

At the same time, its pure-play foundry model gained traction. With no competing products of its own, TSMC became the "Switzerland of semiconductors", a safe, high-quality partner for everyone. Fabless giants like Qualcomm, Broadcom, MediaTek, and Nvidia, which started working with TSMC in the late 1990s, turned to the company to bring their designs to life.

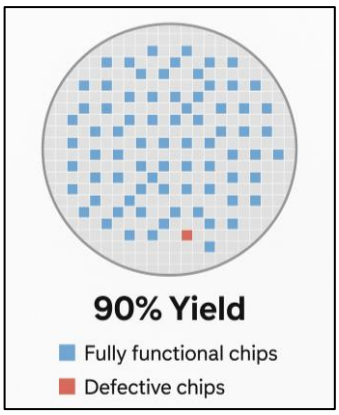
By the late 2000s, It had grown from a single fab into the world's largest dedicated semiconductor foundry, outpacing rivals like UMC, SMIC, and Chartered Semiconductor. Its customer base expanded from niche designers to some of the most powerful chipmakers in the world. And while Intel still led in cutting-edge process nodes, TSMC was catching up fast.

Before we move ahead, lets understand some technical terms that would help us understand what was actually happening in the semiconductor chips and how TSMC changed the game through technical breakthroughs:

Side Note: The Technical Know-How

Yield - To understand how chips are made, picture a wafer, a thin, round slice of pure silicon. On each wafer, hundreds (sometimes thousands) of individual chips are printed. But not all of them come out perfectly. That's where yield comes in. Yield is the percentage of usable, functional chips produced on a single wafer.

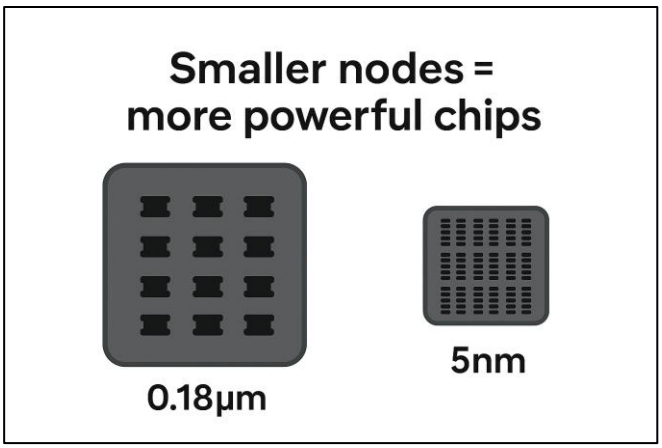
Let's assume if 500 chips are printed and 450 work as intended, that's a 90% yield. High yield means lower cost, better efficiency, and reliable delivery, especially critical when you're producing millions of chips on tight launch timelines. TSMC's reputation was built on its ability to deliver high yields even at cutting-edge nodes, and that consistency is one of the biggest reasons it became the go-to manufacturer for clients like Apple, Nvidia, and Qualcomm (more on this later).



(The circular shape is a silicon wafer, with each square representing a chip. Blue squares are functional dies; red ones are defective.)

TSMC, The Titan | Semiconductor Series

Nodes Size - In the semiconductor world, you'll often hear terms like "5nm" or "0.18µm". These refer to the process node, or the scale of the smallest features that can be etched onto a chip. The units are micrometers (µm) and nanometers (nm). For context, 1µm is one-millionth of a meter, and 1nm is one-billionth. The smaller the node, the more transistors you can fit on a chip, which typically means better performance, lower power consumption, and more efficient use of space. So when TSMC advanced from 0.35µm to 0.18µm, and eventually to 5nm and below, it wasn't just an engineering flex, it enabled everything from longer-lasting phone batteries to faster AI chips.



Now, let's continue on the rise of TSMC and how the next decade of 2010s, where it didn't just close the gap, it sprinted ahead.

From Steady Hand to Global Powerhouse

By the early 2000s, TSMC was hitting its stride. It had proven the viability of the pure-play foundry model. Fabless chip companies were growing rapidly, and TSMC was becoming their most trusted manufacturing partner. It had mastered the sub-micron era, rolled out copper interconnects at 0.13µm, and was preparing to enter the deep-submicron world with 90nm and 65nm nodes.

Morris Chang, then in his early 70s, had spent nearly two decades building the company from scratch. Confident in the leadership bench he'd developed, he stepped down as CEO in 2005, staying on as chairman. The company's growth continued under new management, and Chang fully retired from day-to-day operations in June 2009.

But just as the next phase of the semiconductor boom was picking up, TSMC began to stumble.

TSMC, The Titan | Semiconductor Series

The 40nm Setback and Nvidia's Warning

Around 2009, TSMC was ramping its 40nm node, but ran into yield issues. These issues hit Nvidia particularly hard. The company's GPUs, produced on TSMC's 40nm process, faced delays and performance problems, leading to a public fallout. Nvidia even threatened to shift its orders to GlobalFoundries. For a foundry whose business depended entirely on customer trust, this was a red flag. The once-smooth execution engine had misfired at a critical juncture.

In response, the board turned back to the one person who had built the company's foundation, Morris Chang. And so, on June 12, 2009, the same day TSMC had planned to announce his departure, it instead announced his return as CEO, with Chang resuming day-to-day leadership while continuing as Chairman.

He immediately steered the company back to its strengths: process discipline, customer alignment, and long-term technological bets. He increased capital spending, strengthened R&D, and made bold moves toward the next frontier in chipmaking: FinFET transistors and EUV lithography.

Enter Smartphones and Apple

As TSMC got back on track, the world was undergoing a major technological shift. The smartphone era was taking off since Apple's first iPhone. These new age devices needed chips that were smaller, faster, and dramatically more power-efficient than traditional PC processors.

This trend played directly into TSMC's hands. It had no legacy desktop business to protect. It simply became the best at making small, efficient chips for mobile computing.

Then came the inflection point: Apple.

In 2013, Apple dropped Samsung as its chip supplier and handed exclusive manufacturing of its A-series mobile chips (used in iPhone) to TSMC. That relationship grew into one of the most strategically important deals in modern tech.

TSMC gave Apple:

First access to new nodes like 16nm, 10nm, 7nm, and later 5nm.

A customized production roadmap tailored to Apple's high-performance, low-power needs.

Reportedly, a per-chip pricing model, unlike the per-wafer standard applied to other clients.

Apple, in turn, became TSMC's largest customer, accounting for ~25% of revenue in later years. This massive, predictable demand helped TSMC scale its most advanced nodes faster than anyone else.

TSMC, The Titan | Semiconductor Series

It wasn't just Apple, other major chipmakers like Qualcomm, MediaTek, and Nvidia also began relying on TSMC to manufacture their chips. They were drawn by TSMC's cutting-edge technology, reliable output, and promise not to compete with customers. As smartphones became the main computing device for billions of people, these partnerships helped TSMC scale up like never before, placing it at the center of the mobile tech boom. To keep up with growing demand, TSMC rapidly expanded its manufacturing capacity, building new factories and investing heavily in R&D.

And while TSMC was scaling new heights, Intel, the former champion of semiconductor manufacturing, was falling behind, plagued by delays and missed node transitions.

(TSMC's first chip for Apple was the A8—a 20nm processor and the first 20nm chip to be mass-produced at scale.)

The Node Race and Intel Falls Behind

From 2015 onward, TSMC began to pull away from the competition.

- It launched its 16nm FinFET process in 2015.
- Followed by 10nm in 2017, and 7nm EUV in 2018.
- By 2020, it had begun volume production of 5nm chips, with 3nm close behind.

Meanwhile, Intel, once the undisputed leader in process technology, was falling behind.

- Its 10nm node was delayed by nearly 4 years.
- It struggled with yield and complexity.
- And its vertically integrated structure made it harder to adapt quickly to changing markets.

TSMC, The Titan | Semiconductor Series

TSMC, in contrast, had a laser-focused mission: manufacture chips for others, no product lines, no design ambitions, just execution at scale.

By the late 2010s, TSMC had overtaken Intel in process leadership, and even Intel began outsourcing parts of its chip production to TSMC, a symbolic shift in industry power.

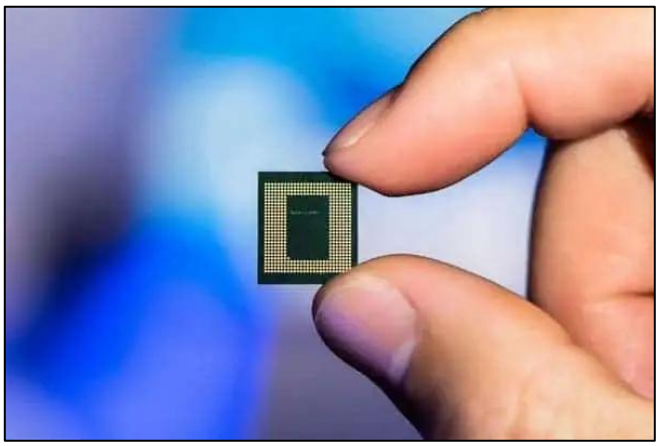
Becoming the World's Foundry

By the early 2020s, TSMC had become the backbone of the global electronics industry. It was producing over 90% of the world's advanced chips (under 7nm), with a client list that included Apple, Nvidia, AMD, Qualcomm, Broadcom, MediaTek, and even Intel. Its 3nm EUV node had entered production, and 2nm with gate-all-around (GAA) transistors was already in development.

It wasn't just making chips, was enabling entire industries:

- AI (through Nvidia)
- iPhones and Macs (through Apple)
- Data centers and gaming (through AMD)
- 5G and connectivity (through Qualcomm and MediaTek)

Even Samsung, a direct competitor that traditionally both designed and manufactured its own Exynos chips, made a major shift: for its 2023 flagship Galaxy S23 and all flagship model after that, it abandoned Exynos globally and adopted Qualcomm's Snapdragon 8 Gen 2, a chip designed by Qualcomm and manufactured by TSMC on its 4nm node. This move helped Qualcomm capture nearly 40% of the global smartphone SoC market, most of it riding on TSMC's manufacturing leadership.



(Source: GizChina, a 3nm chip by TSMC)

TSMC, The Titan | Semiconductor Series

TSMC Today and the Road Ahead

As of 2024, TSMC isn't just the world's most important chip manufacturer, it's the foundation layer of the modern digital economy. The company controls an estimated 60–65% of the global foundry market, and an even more astonishing 90%+ of all advanced chip manufacturing (below 7nm). From your iPhone and MacBook to the chips training the latest AI models and powering data centers, TSMC is the invisible engine behind it all.

Technologically, TSMC is miles ahead of its competitors. It is the only company in high-volume production at the 3nm EUV node, and already deep into 2nm development using Gate-All-Around (GAA) transistor architecture, which promises even greater performance and power efficiency. No other foundry, not even Samsung or Intel has reached this level of scale or consistency at these advanced nodes. Intel, for example, is still working toward catching up with its Intel 3 and 20A process nodes, while Samsung struggles with yield and client retention.

TSMC's client list reads like a who's who of the global tech economy:

- Apple (its largest customer, exclusive for 3nm)
- Nvidia (AI and GPU chips)
- AMD (CPUs and GPUs)
- Qualcomm (Snapdragon mobile SoCs)
- MediaTek, Broadcom, and even Intel (for outsourced manufacturing of select designs)

(And this is just the tip of the ice berg) This dominance has raised a natural question: If TSMC is already this powerful, where does it go from here? And perhaps more importantly: Can anyone else catch up?

The answer lies in how TSMC is future-proofing itself, not just through better chips, but through better foresight.

It is:

- **Scaling globally**, with advanced fabs under construction in Arizona, Japan, and Germany, to reduce geopolitical risk and serve key markets locally.
- **Investing in next-generation packaging technologies** like CoWoS, SoIC, and InFO, which allow chips to be stacked and integrated for even higher performance without always needing smaller nodes.
- **Doubling down on A**I, becoming the sole foundry partner for Nvidia's H100 and upcoming B100 chips, vital for generative AI workloads.
- **Preparing for 2nm and beyond**, with early-stage R&D on new materials and transistor structures like nanosheets and possibly 2D materials down the road.

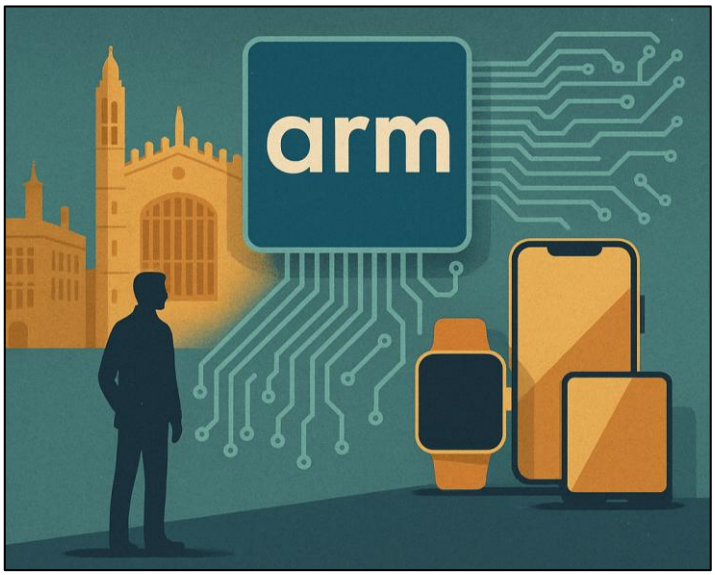
TSMC, The Titan | Semiconductor Series

The reality is: as long as the world runs on silicon and we still do, no other company is structurally positioned to catch TSMC. The capital intensity, execution discipline, ecosystem partnerships, and decades of node leadership are not something a competitor can simply “build” overnight. The moat is wide, and it's deep.

TSMC doesn’t sell devices. It doesn’t chase headlines. But it may be the most quietly indispensable company of the 21st century, powering everything, from smartphones to supercomputers, and now, the future of AI.

The Role of ARM | Semiconductor Series

Advanced RISC Machines



In this chapter we are going to talk about **ARM Holdings**, a chip blueprint design company that took the world by the storm, changed the entire semiconductor value chain, and became the industry standard that powers the majority of smart portable devices that we use today and will probably use in the future.

Cambridge Roots and the Birth of an Unusual Idea (1980s)

In the early 1980s, while the likes of Apple and IBM were leading the personal computing charge in the United States, a small British company named Acorn Computers was quietly building something revolutionary in Cambridge. Tasked with developing the BBC Micro as part of a national computer literacy project, Acorn succeeded in placing computers into thousands of UK homes and classrooms. But inside these machines were processors built by others, first Motorola, then Intel chips that were powerful, but far too energy-hungry to be used in anything portable.

Acorn's engineering duo, Sophie Wilson and Steve Furber, saw an opportunity. What if a processor didn't need to be big, hot, and power-thirsty? What if it could be small, efficient, and built from scratch for lean performance? Embracing a design philosophy known as Reduced Instruction Set Computing (RISC), they stripped away the unnecessary complexities of traditional chip architectures and focused on doing fewer things, but doing them exceptionally well. The result was ARM1, a processor with just ~25,000 transistors that consumed a fraction of the power of its contemporaries. It wasn't powerful enough to run full desktop applications, and it lacked compatibility with the dominant Intel and Motorola software ecosystems. But it worked. Most importantly, wasn't power hungry.

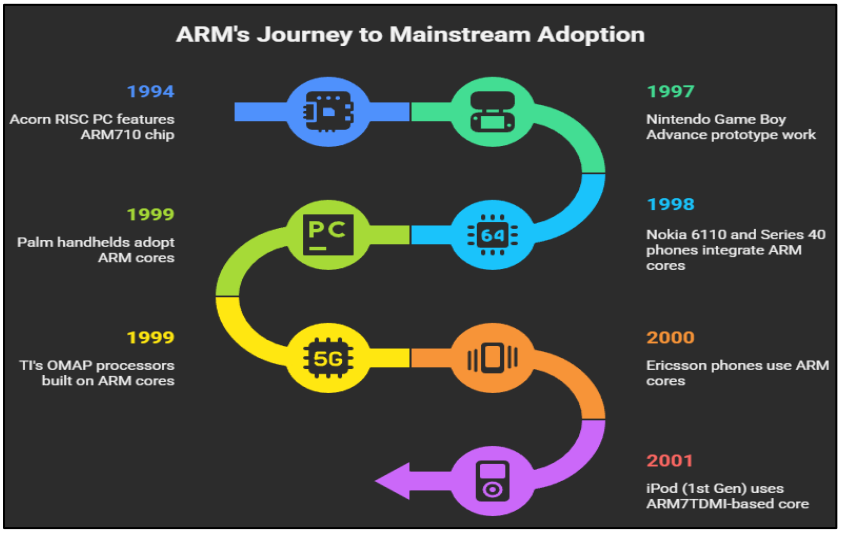
The Role of ARM | Semiconductor Series

The Formation of ARM and a New Business Model (1990s)

This tiny chip became the foundation of something much bigger. In 1990, with Apple planning a new handheld device called the Newton, it partnered with Acorn and VLSI Technology to spin out a new company: Advanced RISC Machines, ARM for short. From the outset, ARM defied convention. It didn't manufacture chips. It designed them and licensed those designs to others. With just 12 engineers at the start, ARM focused on one thing: building efficient processors for portable, battery-powered devices.

The Apple Newton, powered by the ARM610 chip, launched in 1993. Though the Newton failed commercially, it validated the potential of ARM's architecture. What could have been a dead end became a proof of concept. Over the next decade, ARM's processors quietly became the go-to choice for embedded electronics and handheld devices from PDAs and feature phones to early gaming consoles and digital cameras.

Key customers and devices that adopted ARM (before smartphones)



By the time smartphones emerged in the mid-2000s, ARM was everywhere. Its energy-efficient chips were inside millions of portable devices well before the category exploded.

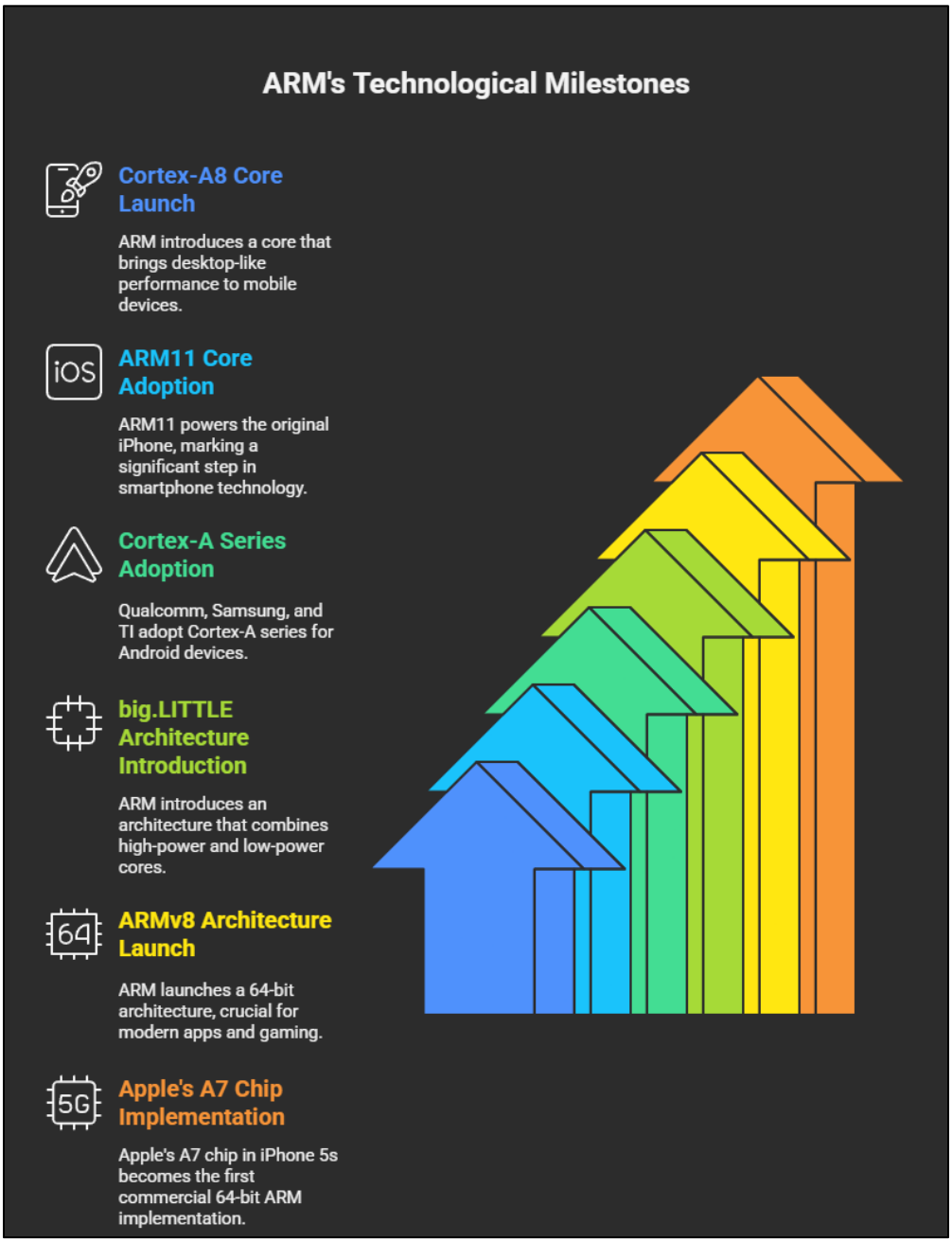
From the Shadows to the Spotlight: ARM's Rise in the Smartphone Era

By the early 2000s, the world was on the cusp of a mobile revolution. Internet access was going wireless, phones were turning smarter, and consumers were expecting more computing power in their pockets. Devices needed to be fast, connected, and most importantly energy-efficient. It was the perfect storm for ARM.

The Role of ARM | Semiconductor Series

Having already made quiet inroads into feature phones and PDAs in the late ’90s, ARM entered the 2000s with a key advantage: its chips were lightweight, power-efficient, and scalable, everything smartphone makers needed to build the next generation of mobile devices.

Then in 2000s, the company made inroads big-time in terms of tech that helped it achieved some major milestones.



The Role of ARM | Semiconductor Series

Throughout 2000s-2010s, ARM's processor designs became near-universal across mobile and steadily expanded into adjacent categories. Additionally, smartphone companies also started designing their own chips using ARM's architecture license, allowing deep customization.

Apple used custom ARM-based chips in all iPhones and iPads.

Qualcomm dominated Android with Snapdragon processors was built on ARM cores.

Samsung produced ARM-based Exynos chips for its flagship Galaxy phones.

Nvidia built its Tegra line of mobile processors on ARM, used in tablets and gaming consoles.

MediaTek and **Spreadtrum** built low-cost ARM-based chips for emerging markets.

Interestingly, the story does'nt end with phones and tablets. Today, ARM processors are being embedded in Smart TVs (e.g., LG, Samsung), Wearables and fitness trackers (Fitbit, Android Wear), Consumer electronics like printers, routers, and game consoles, Automobiles, handling infotainment systems, sensor control, and early autonomous functions, IoT devices, from smart thermostats to industrial sensors, etc.

By 2015, over 15 Bn ARM-based chips were being shipped annually. It had gone from a niche UK-based licensing firm to the world's most prolific chip architecture company, found in over 95% of smartphones and billions of other connected devices.

What Made Customers Stick? ARM's Moats

ARM wasn't the only chip designer in the world but it built a combination of following technical and non-technical moats that made it nearly impossible to displace.

Licensing Flexibility: Customers could choose to license standard ARM cores or go deeper by licensing the ARM architecture itself, giving them freedom to customize, differentiate, and control their supply chains.

Power Efficiency Focus: In an increasingly mobile-first world, energy efficiency is more important than raw speed and ARM's designs are light years ahead in that department.

Developer Ecosystem: ARM invested early in building tools, documentation, and partnerships with software developers, making it easy to write apps, drivers, and firmware that "just worked" on ARM chips.

Intel's Missed Shot: How ARM Won the Smartphone War?

While ARM was quietly embedding itself into billions of devices, Intel, the king of PC processors, watched the smartphone boom from the sidelines. For decades, Intel's x86 architecture had dominated desktops and laptops. But as the world shifted toward mobile computing in the 2000s. Unfortunalety, Intel failed to adapt.

The Role of ARM | Semiconductor Series

Its processors, though powerful, were built for performance, not power efficiency. Attempts to shrink desktop-class chips into smartphones led to thermal issues and poor battery life, which was a non-starter for handheld devices. Intel did launch Atom, a line of low-power chips aimed at smartphones and tablets, but it came too late and couldn't match ARM's efficiency or ecosystem adoption.

- Between 2012 and 2015, Intel reportedly spent over $10 Bn subsidizing Atom chips to win mobile market share, including partnerships with device makers like Asus and Lenovo, but with little success.
- By 2015, ARM-based chips were powering over 95% of all smartphones, while Intel held less than 1% of that market.

ARM's architecture was flourishing because it had been purpose-built for mobile from the start. Its lightweight design, customizable licensing model, and massive software support made it the go-to choice for every major smartphone chipmaker: Apple, Qualcomm, Samsung, MediaTek, Huawei, and others.

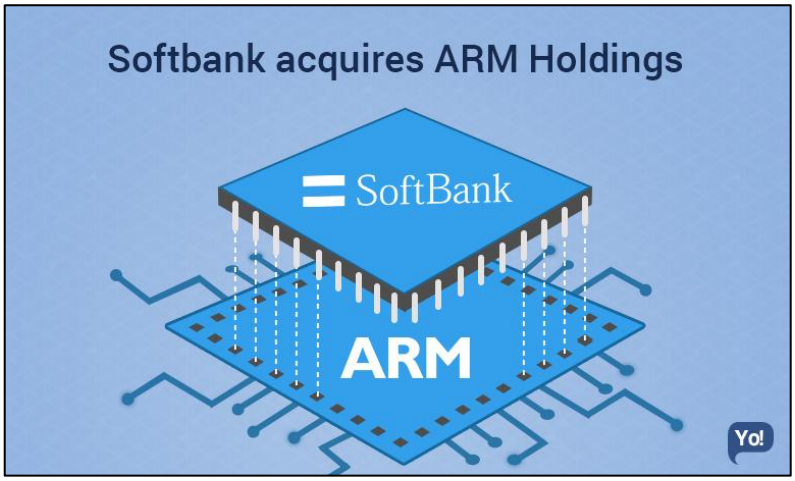
ARM vs Intel: The Smartphone Era Scorecard

Metric	ARM	Intel
Smartphone Market Share	~95%+ of all smartphones	<1%
Core Architecture	ARMv7 / ARMv8 (RISC-based)	x86 (CISC-based)
Power Efficiency	Designed for low power from start	Legacy PC design, power-hungry
Business Model	Licensing & royalties (fabless)	Sells physical chips (integrated)
Key Customers	Apple, Qualcomm, Samsung, MediaTek	Few Android vendors (e.g. Asus)
Mobile Chip Investment (2012–15)	Minimal capex, IP-led growth	>$10 billion in mobile chip subsidies
Mobile SoC Division (2016)	Expanding	Shut down

The Role of ARM | Semiconductor Series

In 2016, Intel quietly shut down its mobile SoC division, effectively exiting the smartphone race. As mobile became the dominant form of computing, ARM became the architecture of the future, and Intel, once synonymous with chips, faded from the mobile conversation entirely.

Acquisition by SoftBank and a Broader Vision (2016–2020)



That same year (2016), ARM became the center of a different kind of story. Japanese conglomerate SoftBank, led by Masayoshi Son, acquired the company for $32 billion in one of the largest tech deals of the decade. Son wasn't just buying a chip designer. He believed ARM would power the next era of computing, across IoT, AI, robotics, and cloud infrastructure. Taking ARM private gave it the flexibility to invest in longer-term bets without the short-term pressure of quarterly earnings.

ARM used this breathing room well. It expanded into new markets, introduced the Neoverse line for cloud data centers, launched the Ethos AI processor family, and developed the Cortex-X series for premium smartphones. Even though the company operated outside the public spotlight, it continued to dominate mobile and edge computing, with its architecture spreading to cars, smart homes, and industrial devices.

The NVIDIA Deal that Wasn't (2020–2022)

As you know, SoftBank is not an (Not Berkshire ;) holding company. It is a venture investor. And exits are always part of the plan.

The Role of ARM | Semiconductor Series

In 2020, SoftBank announced a proposed $40 billion sale of ARM to NVIDIA. The deal promised to merge ARM's universal CPU ecosystem with NVIDIA's graphics and AI acceleration prowess. But this move immediately triggered regulatory alarms across the globe. Governments saw ARM not just as a business but as critical infrastructure. In the UK, it was a matter of national pride. In the US and EU, it raised antitrust concerns. Additionally, in China, the optics of an American company taking over such a vital component of the global tech stack added fuel to an already tense geopolitical standoff.

After nearly 18 months of investigations, lobbying, and delays, the NVIDIA-ARM deal collapsed in 2022. SoftBank pivoted. It restructured ARM, focused it more sharply on AI and cloud applications, and prepared it for a public offering.

ARM Returns to the Public Markets (2023)

By September 2023, ARM returned to the markets with a NASDAQ IPO that valued the company at over $65 billion. For SoftBank, it was about unlocking value but for ARM, it was a public reintroduction, not just as a mobile chip licensor, but as the foundation of next-generation computing.

What made the entire saga remarkable was not just the corporate drama, but the consistency of execution throughout. While executives and lawyers debated ownership, ARM engineers kept shipping. It continued powering 99% of smartphones. It struck new deals with Amazon for Graviton processors and Tesla for its Full Self Driving chips. It became increasingly relevant in cloud data centers and edge AI, where its power efficiency and flexible architecture gave it an edge over legacy incumbents.

A Business Model that Scales as Customers Scale

ARM operates on a licensing-and-royalty model which scales beautifully with the success of its customers. Today, ARM makes money by licenses its chip designs or architecture to companies like Apple, Qualcomm, Samsung, MediaTek, and many others. ARM mostly earns from companies in two ways:

- Customers pay ARM an upfront fee to access processor core designs or license the entire ARM instruction set architecture (ISA) to design custom chips (what Apple does). This licensing fee is usually a multi-million dollar contract, paid once or over a few years.
- Once a customer starts manufacturing and selling chips based on ARM's design, ARM earns a royalty on each chip shipped. This is typically a small percentage of the chip's price, often between 1 to 2 percent. So yes, when Apple sells an iPhone with an ARM-based A-series chip inside, ARM gets a royalty for each unit. Same goes for chips sold by Qualcomm (Snapdragon), Samsung (Exynos), MediaTek, and others. If it uses ARM's cores or architecture, ARM earns a slice.

The Role of ARM | Semiconductor Series

Where ARM Stands Today? The Silent Standard of Modern Electronics

As of today, ARM is no longer just a mobile-first company. It is the default architecture across nearly every category of electronics, and its dominance is not symbolic, it is backed by scale.ARM-based chips power over 99% of smartphones, 80% of digital cameras, 95% of wearable devices, and over 60% of smart TVs and home appliances.

- In automotive, ARM powers infotainment systems, driver-assistance features, and in many cases, sensor fusion chips used in ADAS (Advanced Driver Assistance Systems).
- In data centers, ARM is making rapid progress. Amazon's Graviton processors, built on custom ARM architecture, are now powering a significant portion of AWS workloads, offering better performance per watt than traditional Intel Xeon-based servers.
- In AI and edge computing, ARM's low power and scalable designs are being used in smart cameras, industrial sensors, and on-device inferencing for vision and language tasks.

ARM's competitive edge lies in its DNA. Unlike Intel, which designed chips originally for desktops and later tried to adapt them to other devices, ARM's architecture started with power-sensitive use cases. That advantage has compounded over decades. ARM designs are modular, license-friendly, and optimized for customization, all of which allowed it to scale across device categories and geographies.

Its ecosystem is another moat. Tens of thousands of developers are trained on ARM's tools. Device makers, operating systems, and software platforms, from Android to embedded Linux, are tightly integrated with ARM.

In a world that is going from billions of devices to trillions, where every toaster, drone, and satellite is essentially a computer, ARM is not just relevant. It is foundational.

Closing Thoughts

Today, ARM is the default computing architecture across consumer electronics, automotive systems, and increasingly, the data center. Its chips run over 250 billion devices worldwide. They live inside thermostats and smartphones, drones and satellites. Its software ecosystem spans Android, Linux, and custom real-time operating systems. And its business model, licensing rather than manufacturing remains as scalable and capital-efficient as ever.

ARM's success was never built on hype. It was built on insight: that the future would not belong to the biggest chip, but to the smartest one. At the cost of repeating, in today's fast moving world, ARM's role is not just important. It is foundational.

While others battled for control over fabs, supply chains, and chip inventories, ARM sold the blueprints. It let the world build on top of its architecture, and in doing so, embedded itself into everything from your phone to your thermostat and may be even to the cloud server routing this very newsletter.

Fabless Companies | Semiconductor Series

Fabless, Fearless, and Worth Billions

What are FABLESS Companies?

Simply put, fabless companies are those that design semiconductor chips but do not manufacture them. Instead, they outsource the manufacturing part to companies called foundries, such as TSMC or others, which specialise in manufacturing chips.

Think of them like a fashion designer who comes up with the design of clothes (style, cut, fabric choice), but doesn't own a sewing factory. They send their designs to a third-party clothing manufacturer who actually stitches and produces the clothes. That's exactly how fabless chip companies work. Some prominent names include:

Company	Headquarters	Known For
Qualcomm	USA	Mobile processors (Snapdragon), 5G modems
Nvidia	USA	Graphics cards (GPUs), AI chips (like H100)
Broadcom	USA	Networking, Wi-Fi, Bluetooth, and storage chips
AMD	USA	CPUs (Ryzen), GPUs (Radeon), server chips (EPYC)
MediaTek	Taiwan	Smartphone SoCs (Helio, Dimensity), smart TVs

Fabless Companies | Semiconductor Series

If you widen the lens a bit, even Apple fits the fabless model. It designs its chips through its in-house silicon team, while the actual manufacturing is handled exclusively by TSMC. The difference? Unlike Nvidia or Qualcomm, Apple doesn't sell its chips to anyone else. Instead, it builds them solely for its hardware — iPhones, Macs, Watches, and more.

Market Darlings

Unless you've been living under a rock, you've probably noticed the meteoric rise in the market capitalisation of publicly traded companies over the past few years. In fact, a few of the names we just mentioned have now crossed the trillion-dollar mark in market cap, a staggering feat for companies that don't manufacture a single chip themselves. Here is a quick summary:

Company	Market Capitalization	Last 5Y CAGR (%)
Nvidia Corp	$4.1 Trillion	~70%
Apple Inc	$3.1 Trillion	~13%
Broadcom Inc	$1.3 Trillion	~75%
Qualcomm Incorporated	$166 Billion	~7%
AMD Inc	$260 Billion	~12%

(Market Capitalization Number is Slightly Dated)

To put that in perspective, as of 2024, only 19 countries had a GDP above $1 Trillion. On a lighter note, if Nvidia, Apple, or Broadcom were countries, their market caps would earn them a seat at the G20 table ;)

Product Companies at Heart

Although, from the outset, it appears that fabless companies are merely chip designers, upon closer examination, there is much more to these companies than just chip design. If you step back and observe, at their core, these companies are product companies that don't own manufacturing facilities.

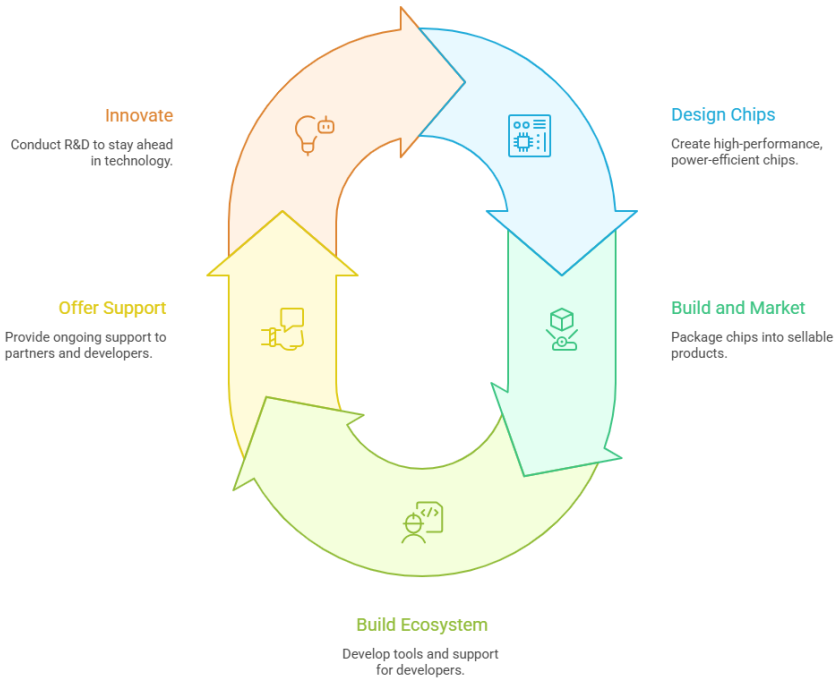
Fabless companies don't only sell anonymous silicon; they also brand their chips (Snapdragon, GeForce, Rayzen, etc.) and enjoy strong brand recall (this phone has a Snapdragon 8 Gen3!). Here are some examples:

Fabless Companies | Semiconductor Series

Fabless Company	Flagship Products / Platforms
Nvidia	GeForce (gaming GPUs), RTX (ray tracing), CUDA (AI compute), H100/Blackwell (AI chips), Nvidia Drive (autonomous vehicles)
Qualcomm	Snapdragon (mobile SoCs), Snapdragon Modem, Snapdragon Sound, Snapdragon X Elite (PC chips), RF front-end platforms
AMD	Ryzen (desktop CPUs), EPYC (server CPUs), Radeon (GPUs), Xilinx FPGAs, ROCm (AI/ML software stack)
MediaTek	Dimensity (premium 5G SoCs), Helio (midrange chips), Pentonic (smart TV chips), Kompanio (Chromebooks)
Broadcom	NetXtreme (Ethernet), Wi-Fi 7/BT combo chips, Tomahawk (switching ASICs), Jericho (routing chips), custom AI accelerators for Google
Apple (fabless-like)	M1/M2/M3 (Mac), A17 (iPhone), H-series (AirPods), S-series (Watch) — all built by TSMC, but Apple-designed

A Full Stack Playbook – Key to the success of fabless companies

If we have an overview, running a successful fabless company is about effectively managing the following FIVE steps:



Fabless Companies | Semiconductor Series

Step 1, Chip Design à This is where it all begins, designing high-performance, power-efficient chips tailored for specific use cases. Companies often use a mix of third-party IP (e.g., ARM cores, DSPs) and proprietary blocks to design chips.

Step 2, Build and Market the Product à A chip is useless unless it's packaged as a product. This is where design becomes a sellable platform (e.g., Snapdragon, GeForce, Ryzen, etc). This step turns engineering into revenue.

Step 3, Ring fence by building an Ecosystem à This is where their IP gets supercharged. The best fabless companies build development tools, software development kits (SDKs), support libraries, certifications and training programs. These tools attract developers and customers. Companies often try their best to make these tools easy to use, but extremely difficult to let go. Examples include Nvidia's CUDA, Qualcomm's Snapdragon SDKs, and Apple's tight integration with iOS/macOS, among others. This is what creates stickiness (more on this below). Even if someone builds a better chip, switching is painful.

Step 4, Offer Support à Fabless companies don't act like a vendor but a partner. They provide ongoing support to OEMs, integrate chips, and assist developers with documentation and forums, while also partnering in co-designing and optimisation. For example, Qualcomm helps phone brands integrate chips across hundreds of SKUs and regions. Similarly, NVIDIA works with cloud providers to optimise AI workloads. This deepens relationships and helps win repeat business.

Step 5, Keep building Cutting-Edge IP à In this business, if you're not inventing, you're losing. fabless companies future-proof their edge by doing R&D into new process nodes, packaging, or vertical capabilities. In the past, some companies have also acquired others to stay ahead of the curve (AMD → Xilinx, Nvidia → Mellanox). The idea is to stay ahead of the curve because chip cycles are brutal and fast.

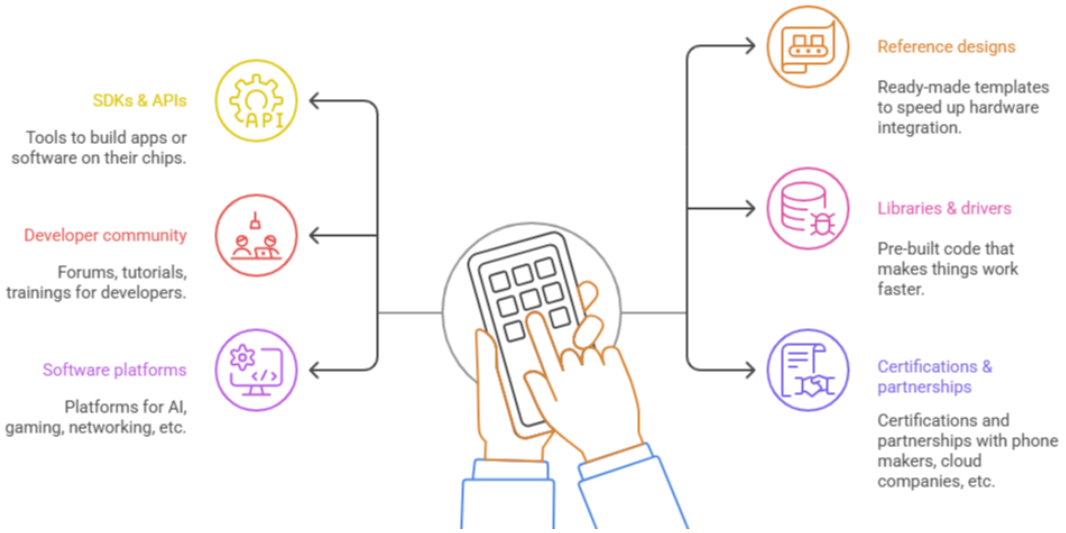
In short, a successful fabless player has a full-stack playbook where each step compounds the previous one. Design gets you in the door, but product, ecosystem, support, and innovation keep you in the game, and miles ahead of commoditised players.

The Moat – You don't fall in love with the chip, but the experience it enables

If you have read the above step with absolute precision, one thing must have become extremely clear: Ecosystem is the real moat that fabless companies have created around themselves. Now, before getting into the nuances of ecosystems, let's first define an ecosystem of a fabless company.

Ingredients of an ecosystem

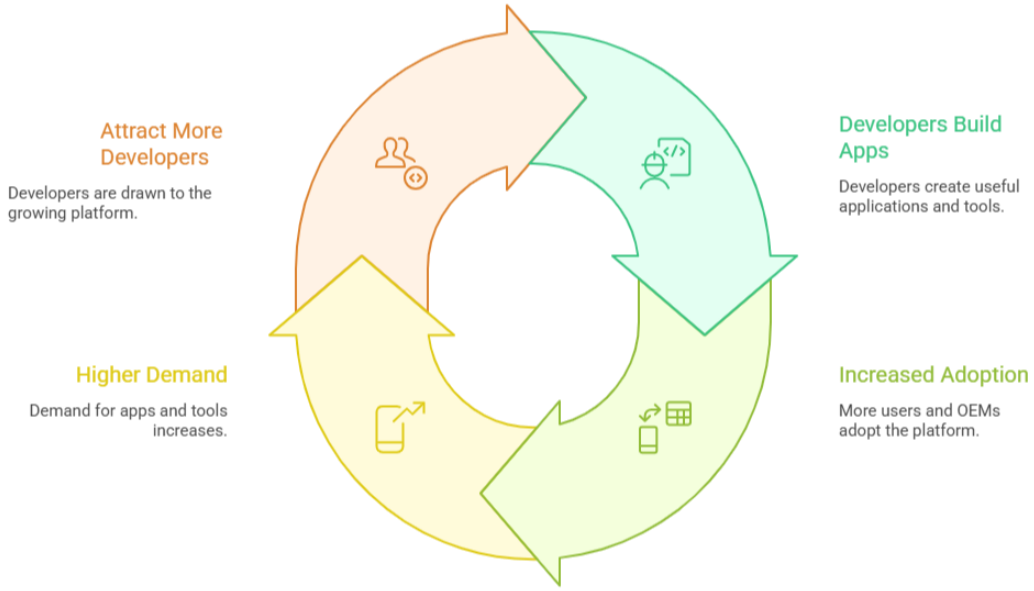
An ecosystem is everything a company builds around the chip to help customers, developers, and partners use it easily and powerfully. It includes: Software Development Kits (SDKs) & APIs, Reference designs, Developer community, libraries and drivers, Software platforms, certifications, and partnerships, etc.



The step where MAGIC happens

Holistically seeing, this is the step where the chip stops being "Just a Part" and becomes a complete solution. At this stage, it's not just about the chip; it's about the whole experience built around using it. This is because of the following reasons: a) an ecosystem makes life easy for engineers and developers, b) an ecosystem unlocks the full power of the chip, and c) an ecosystem creates a “sticky” platform that keeps customers coming back. In summary, with an ecosystem, a chip becomes a platform everyone wants to build on.

Network Effect at Play



Fabless Companies | Semiconductor Series

Giving up an Ecosystem is Extremely Difficult

Once a developer, partner, or customer becomes part of the ecosystem, it becomes difficult to leave the ecosystem, even if a rival product is developed. This happens because of very high switching costs because switching the ecosystem means Rewriting code, Relearning tools, Retesting everything, Retraining people, and, most riskily, risking bugs, delays, and performance issues. Imagine having to relocate your entire office to a new city just because the rent is slightly cheaper. Most won't do it unless the benefit is enormous. So even if a rival chip is 10% better, it's often not worth the hassle of leaving a mature ecosystem.

Real Ecosystems at Play – Examplesof Nvidia and Qualcomm

Ecosystem Components	Nvidia (AI & GPU)	Qualcomm (Mobile & Modem)
Software Development Kits	CUDA SDK: Developers build AI and ML models using Nvidia's tools.	Snapdragon SDK: Helps phone makers integrate camera, AI, and audio features.
Support Libraries and Tools	cuDNN, TensorRT: Libraries that make models faster.	RF front-end tools: For 4G/5G antenna tuning. Reference designs: OEMs can launch phones faster.
Events, Certifications and Partnerships	Nvidia GTC: Conferences and training for developers. Partnered with cloud giants (AWS, Azure) — so you can train your model anywhere using Nvidia.	Carrier certifications: Already pre-certified with global mobile networks.
Switching Costs	Once you build AI models in CUDA, it's very hard to shift to AMD or Intel — you'd have to recode everything.	Switching to another chip means delays, carrier approvals, and lost optimization.

Some Probable Reasons - Why Fabless Companies get a high PE?

Now, we come to the most interesting aspect, the question with which we started this newsletter at the first place. It is essential to understand that there can be numerous answers to this abstract question, and all of which can be a probable explanation. The following are the few factors that we believe fetch a higher multiple to fabless companies:

Fabless Companies | Semiconductor Series

1. This is an IP-driven, scalable business

Fabless firms own the chip design, not the factory. Once the design is complete, it can be manufactured in millions at nearly zero incremental cost. Margins stay high, and scaling doesn't require more physical assets.

2. Brand Recall

Brands like Snapdragon, GeForce, Ryzen, Dimensity have become product signals both to OEMs and end users. This gives fabless companies pricing power and customer pull, even in B2B markets. This kind of brand recall has turned a commodity (chip) into a premium product.

3. The Moat of Ecosystem, Supercharged by Experience

As discussed, Ecosystems include SDKs, reference designs, partnerships, developer support, etc. They create network effects and make switching painful. But most importantly, they create an excellent experience for customers, developers, and integrators. Customers don't just use the chip, they live in the world built around it.

4. Ecosystem enables Repeat Business > Annuity Income

OEMs who integrate a fabless chip into their product and use its tools, dev kits, and certifications rarely switch every cycle. This leads to multi-year design wins, follow-on business, and roadmap loyalty. That lock-in nature converts a project-based business into a sticky on with repeat-revenues.

5. Investors love Annuity Businesses

Predictable cash flows, high gross margins, low churn, and scalable IP = Strong Stability. Add in optionality for growth (AI, Auto, IoT, 5G…), and you get growth + stability, the combination every investor wants in markets.

In summary, Fabless companies command premium valuations because they don't just sell chips, they scale ideas. In the world of semiconductors, it's easy to get lost in talk of factories, machines, and tiny chip sizes. But the real power lies elsewhere, with fabless companies. These aren't just chip designers; they're product creators, brand builders, and masters of experience. The chip is just the engine; the magic is in the smooth, reliable world they build around it. They don't just sell hardware; they sell confidence, loyalty, and something people come back to again and again. And that's why the market rewards them with rich valuation premiums. Because, in an industry powered by circuits, fabless companies have figured out how to build something even more powerful: EXPERIENCE.

Fabless Companies | Semiconductor Series

1. This is an IP-driven, scalable business

Fabless firms own the chip design, not the factory. Once the design is complete, it can be manufactured in millions at nearly zero incremental cost. Margins stay high, and scaling doesn't require more physical assets.

2. Brand Recall

Brands like Snapdragon, GeForce, Ryzen, Dimensity have become product signals both to OEMs and end users. This gives fabless companies pricing power and customer pull, even in B2B markets. This kind of brand recall has turned a commodity (chip) into a premium product.

3. The Moat of Ecosystem, Supercharged by Experience

As discussed, Ecosystems include SDKs, reference designs, partnerships, developer support, etc. They create network effects and make switching painful. But most importantly, they create an excellent experience for customers, developers, and integrators. Customers don't just use the chip, they live in the world built around it.

4. Ecosystem enables Repeat Business > Annuity Income

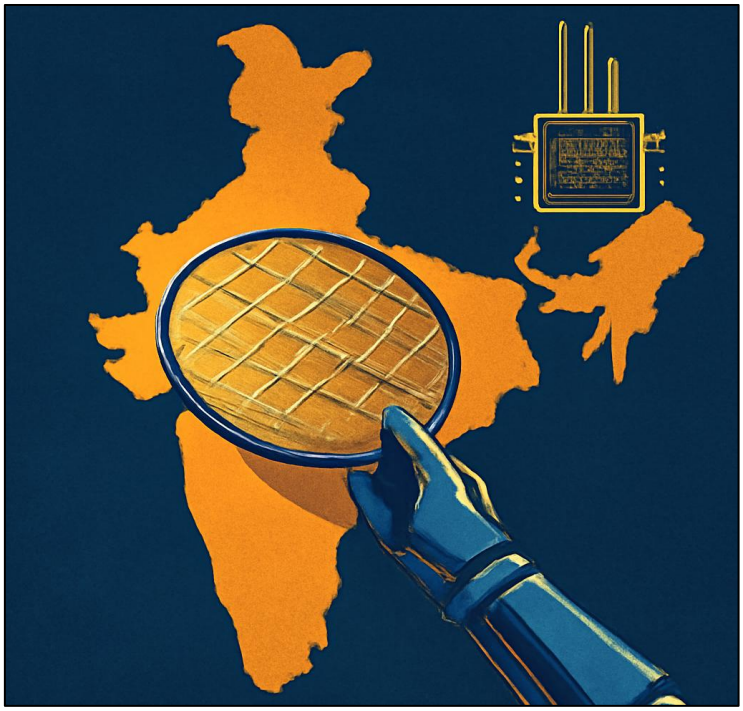
OEMs who integrate a fabless chip into their product and use its tools, dev kits, and certifications rarely switch every cycle. This leads to multi-year design wins, follow-on business, and roadmap loyalty. That lock-in nature converts a project-based business into a sticky on with repeat-revenues.

5. Investors love Annuity Businesses

Predictable cash flows, high gross margins, low churn, and scalable IP = Strong Stability. Add in optionality for growth (AI, Auto, IoT, 5G…), and you get growth + stability, the combination every investor wants in markets.

In summary, Fabless companies command premium valuations because they don't just sell chips, they scale ideas. In the world of semiconductors, it's easy to get lost in talk of factories, machines, and tiny chip sizes. But the real power lies elsewhere, with fabless companies. These aren't just chip designers; they're product creators, brand builders, and masters of experience. The chip is just the engine; the magic is in the smooth, reliable world they build around it. They don't just sell hardware; they sell confidence, loyalty, and something people come back to again and again. And that's why the market rewards them with rich valuation premiums. Because, in an industry powered by circuits, fabless companies have figured out how to build something even more powerful: EXPERIENCE.

India's Leap into the Semiconductor Future



Where India Stands? - Semiconductor Design

As discussed earlier, designing a semiconductor is very much similar to building a house. Before you lay a single brick, you hire an architect to design it. They decide things like, How many rooms you need, Where the doors and windows go, How the plumbing and electricity will run, etc… etc…

India plays an important role…

India is already an important piece in the global semiconductor design. You may be surprised to know that ~20% of the world's chip design engineers are based here. Big players like Qualcomm, Intel, Nvidia, Broadcom, and MediaTek have large R&D and design centers in Bengaluru, Hyderabad, and Noida.

…However, cutting edge R&D (mostly) takes place in the USA

If we try to breakdown the division of semiconductor design work between India and the USA (where fabless giants like Nvidia, Qualcomm, Apple, etc are domiciled), this is how the summary looks like:

Developments in India | Semiconductor Series

Engineers in the USA answers questions like, What kind of chip to build?, Who will use it, a smartphone, a server, or a self-driving car?, What features to include?, What performance and power targets to aim for?, how to make it better and faster than the previous model?, etc. Basically, they sketch out the high-level architecture. In addition to technical questions, they also decide the product strategy (when to launch?, whom to sell to?, how to stay ahead of competition?, etc).

On the other hand engineers in India step in once the high level technical and architectural questions are answered. India's engineering teams take over a huge part of the actual building process. Here's what Indian engineers do: Translate the architecture into working logic, Simulate and test the chips, Optimize it for speed and power, Write drivers and firmware to make it work with real devices, Support and fine-tune EDA tools that the entire industry uses.

Though it's not a "boss vs worker" setup but USA defines the "what" and "why" and India executes the "how" with precision and scale.

Chip Manufacturing Giants have bases in India

To get this work done, global fabless companies (like Nvidia) as well as integrated chip majors (like Intel) have established their design centers in India. They do chip design directly (mostly) through their own teams, not through Indian vendors. These are GCCs (Global Capability Centers) which are massive R&D centers, not just back offices. Here's a list of major players with design centers (GCCs) in India:

Company	India Locations	Focus Areas
Intel	Bengaluru, Hyderabad	CPUs (Xeon, Core), SoCs, chipsets
Qualcomm	Hyderabad, Bengaluru	Mobile SoCs, 5G modems, AI engines
Nvidia	Bengaluru, Pune	GPUs, AI chips, verification, CUDA stack
AMD	Hyderabad, Bengaluru	CPUs, GPUs, custom logic blocks
MediaTek	Noida, Bengaluru	Smartphone SoCs, modem design
Texas Instruments	Bengaluru	Analog, embedded processors, DFT, layout
Broadcom	Bengaluru, Hyderabad	Networking chips, custom silicon
ARM	Bengaluru, Noida	Core architecture, verification, firmware
Apple	Hyderabad, Bengaluru	Silicon validation, camera and RF chips
Google	Bengaluru, Hyderabad	Tensor SoCs, TPU firmware

Developments in India | Semiconductor Series

These GCCs Employ 100s to 1000s of engineers who do real design work, especially RTL design (writing the blueprint of the chip), verification, physical design and sometimes contribute to core IP (like modem blocks or AI accelerators) as well. However, the final architecture decisions and strategic IP are still controlled from HQs, typically in the USA or other countries.

Signs of Upward Movement

Encouragingly, things are gradually changing slowly but surely. India is no longer just a "backend verification shop." The directional shift suggests that things are moving toward higher-end architectural design and innovation. Here are some examples:

- **Apple India (Bengaluru)** is reportedly involved in custom silicon design for camera and connectivity modules.
- **AMD's biggest global design center is now in India.** They're expanding leadership roles.
- **Intel India** has played a role in the development of its Xeon and client CPUs.
- **Google and ARM** have expanded local silicon teams with senior leadership presence.

Home Grown Names

In addition to the above, Though India does'nt have a fabless player with similar likes of Qualcomm or Nvidia, there are definately a handful of local early-stage fabless players. Following is the list:

Company	Focus Area
Sankhya Labs	Modem IP, now merged with TATA Elxsi
Saankhya Labs	SDR chipsets (acquired by Tejas Networks)
Incore Semiconductors	RISC-V CPU IP
Morphing Machines	Parallel processing chips (very niche)
Silizium	Automotive-grade MCUs and analog ICs
Mindgrove Technologies	India's first RISC-V based SoC for consumer devices

Developments in India | Semiconductor Series

In addition to the above companies, there are also companies like Trident Techlabs that offer services like ASIC design, PCB/signal-integrity, simulation, and EDA tool support, targeting sectors like defense and telecom. Together, they illuminate how India is progressively building IP, platforms, and services, laying the foundation for a credible domestic semiconductor ecosystem.

Where India Stands? - Semiconductor Manufacturing

This is the stage of actual chip construction. Chip manufacturing happens in an ultra-clean, high-tech factory which in technical terms is called a "foundry". This is the hardest and the most capital-intensive part of the value chain. Though India has never manufactured chips at commercial scale, but for the first time ever, it is now laying the groundwork to build that capability from scratch.

Government Push

In 2021, the Indian government launched the Semicon India Programme (also referred to as Indian Semiconductor Mission (ISM) 1.0) with ~Rs. 76,000 Cr in incentives to attract global manufacturers, build fabs, packaging units (OSAT & ATMP, discussed later in detail), and a local supply chain to reduce reliance on imports. The intial focus is on mature-node fabs (28nm to 65nm), not bleeding-edge nodes like 5nm or 3nm, which are dominated by TSMC and Samsung.

India's First Commercial Fab - Loading…

TATA Electronics + Powerchip Semiconductor Manufacturing (PSMC, Taiwan)

This is expected to be India's first truly commercial foundry (fab) targetted towards making chips for EVs, telecom, industrial, etc uses.

Node: Starting with 28nm and above (mature analog/power chips)

Expected Investment: ~Rs. 91,000 Cr

Status (2025): Construction started; target production in 2026

Though today there is not much to share as far as India's progress in establisihing semicinductor fab is concerned, one thing is clear, India's semiconductor ambitions are taking shape with a strategic focus on mature-node manufacturing, a smart move considering the vast demand for 28nm–65nm chips in sectors like automotive, telecom, industrial electronics, etc. While global giants dominate the cutting-edge node space, India is wisely positioning itself to manufacture mature nodes and filling a big gap.

:

Developments in India | Semiconductor Series

Where India Stands? - Semiconductor Assembly and Testing

After the silicon wafer comes out of the foundry, it now has a lot of miniature brains on a wafer. But you can't plug a wafer into your phone, you need to slice, wrap, and test them before starting to use them.

Building OSAT and Other Capabilities

Though India may not yet boast the presence of large-scale fabs like Taiwan or South Korea, but it's carving out a significant presence in semiconductor assembly and testing (OSAT and ATMP) and other vital supporting materials like chemicals.

Government Push

Establishing OSAT and ATMP facilities is a key component of the GoI's ISM 1.0 program, which offers up to 50% funding for CAPEX. This initiative is designed to make applicant companies highly attractive to investors in OSAT or packaging units. In addition to central government incentives, state governments are offering their own subsidies and incentives to attract investments in semiconductor infrastructure. These policies and incentives vary from state to state.

Some Upcoming OSAT & ATMP Projects

Micron Technology (USA)

Facility: ATMP (Assembly, Testing, Marking and Packing, subset of OSAT) facility in Sanand, Gujarat.

Focus: Import wafers from overseas fabs and package them in India.

Expected Investment: ~Rs. 22,500 Cr (50% from Micron and 50% from Govt)

Status (2025): Construction well underway; expected to start packaging DRAM (Dynamic Random Access Memory) and NAND (Not AND) chips by 2026.

CG Power + Renesas (Japan) + Stars Microelectronics (Thailand)

Facility: OSAT (Outsourced Semiconductor Assembly & Testing) facility near Hyderabad.

Focus: Assembly and testing of microcontrollers and analog/power chips.

Expected Investment: ~Rs. 7,600 Cr

Status (2025): Project recently approved (2024); construction expected to start soon

Developments in India | Semiconductor Series

TATA Electronics

Facility: ATMP (Advanced Packing) facility in Tamil Nadu and Assam.

Focus: Chips for automotive, telecom, and defence

Expected Investment: ~Rs. 7,600 Cr

Status (2025): Pre-construction / under development

Kaynes Technology

Facility: OSAT facility in Sanand, Gujarat.

Focus: Export business

Expected Investment: ~Rs. 3,400 Cr (70-75% funded by Govt - Center + State)

Status (2025): Construction underway. To be ready by the end of 2025

Building Support - Semiconductor Chemicals

You might or might not be aware but indian chemical companies are also taking notable strides in developing a comprehensive ecosystem to support semiconductor manufacturing. Semiconductor manufacturing requires a wide range of specialized chemicals for processes like etching, cleaning and other purposes during the fabrication of chips. India is focusing on the domestic production of semiconductor-grade chemicals to reduce dependency on imports, particularly from countries like Japan and the U.S.

India has several companies that are stepping up to support the semiconductor chemicals manufacturing ecosystem. Some key players include:

Chemical Company	Key Chemicals Produced	Role in Semiconductor Manufacturing
Tata Chemicals	High-purity chemicals, wet etch chemicals, hydrogen fluoride, specialty solvents	**Scaling up production of semiconductor-grade chemicals**, especially for **etching** and **cleaning** stages in fabs.
Deepak Nitrite	Sodium hydroxide, hydrofluoric acid, etching solvents	Produces **etching and cleaning chemicals** critical for **semiconductor wafer fabrication**.
Navin Fluorine	Fluorochemicals (hydrofluoric acid, TFE, specialty fluoropolymers)	Key producer of **fluorochemicals** used in the **etching and cleaning processes** in semiconductor fabs.
Pi Industries	Fine chemicals, custom synthesis chemicals, specialty fluorochemicals	Developing **high-purity chemicals** for **semiconductor production** processes, including specialized chemicals for etching.
SRF Limited	Fluorochemicals, specialty chemicals like hexafluoroethane, refrigerant-grade chemicals	Produces **specialty chemicals** for **semiconductor and electronics industries**.

Developments in India | Semiconductor Series

These foundational efforts in OSAT, ATMP and Chemical Manufacturing are laying the groundwork for a robust semiconductor ecosystem which will equip India to manage and scale fabs when they are established at a larger scale. In the long run, these initiatives will empower India to become a crucial node in the global semiconductor value chain.

Following the footsteps of Taiwan and South Korea

India's semiconductor ambition today is quite similar to where Taiwan and South Korea were in the 1970s–1990s. If we compare India's early efforts today with the early days of Taiwan and South Korea, we find the following similarities:

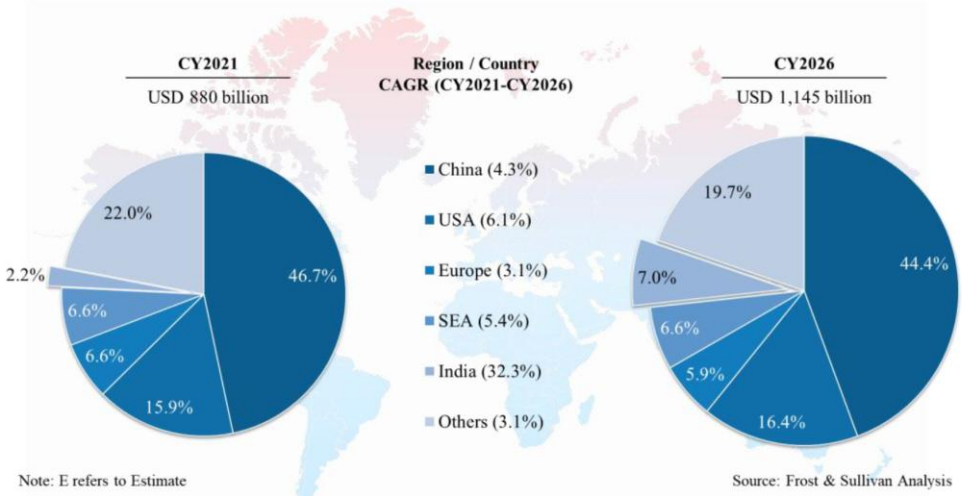
Factor	Taiwan / Korea in Early Days	India Today
Govt-led semiconductor vision	✅ Yes	✅ Yes
Foreign tech partner	✅ Yes (Philips, TI, Micron)	✅ Yes (PSMC, Micron, Renesas)
Start with mature nodes	✅ Yes	✅ Yes (28nm–65nm)
Focus on packaging first	✅ Yes	✅ Yes (Micron, Tata, Renesas JV)
Build ecosystem gradually	✅ Took 10–20 years	🚧 India just starting
National strategic goal	✅ Autonomy + exports	✅ Self-reliance + supply security

India today is where Taiwan was in the 1980s and South Korea was in the early 1990s. The intent, partnerships, and capital are finally in place. Now, it comes down to patience, policy continuity, and execution.

Where India Stands? - System Integration

India's role in system integration within the semiconductor value chain is rapidly evolving, positioning the country at the forefront of global capabilities. With a deep pool of talent and a robust ecosystem, India excels in integrating semiconductor components into high-tech systems for sectors like automotive electronics, telecommunications, consumer devices, and medical technologies. India is already delivering world-class, high-quality integration services. What sets India apart is its ability to match global standards, both in capability and talent, and its swift progression towards scaling these operations to meet the growing demand for advanced semiconductor-based solutions. The following data shows how the global EMS industry is expected to progress, particularly, India:

Developments in India | Semiconductor Series



The China +1 strategy has become a powerful global trend, pushing companies to diversify their supply chains and look for alternatives to China. This has fueled India's growth in system integration, as the country is increasingly seen as a reliable and cost-effective partner in the global supply chain. While China will likely continue to dominate the global system integration market for the foreseeable future, India's significance is only increasing. With its growing infrastructure, talent pool, and competitive advantage, India is well poised to capture a larger share of the global market, positioning itself as a key player in the years to come. Promenent companies include:

- Consumer Sector: Dixon Technologies, VVDN Technologies, Foxconn India, Micromax, PG Electroplast, etc.
- Industrial, Automotive and Other Complex Applications: Siemens India, Cyient DLM, Kaynes Technology, DCX Systems, Avalon Technologies, etc
- Medical: GE Healthcare India, Siemens Healthineers India, BPL Medical Technologies, Philips India, Syrma SGS Tech, etc.

Conclusion

As India continues to carve out its niche in the global semiconductor value chain, its journey is going to be nothing short of remarkable. From designing cutting-edge semiconductor technologies to manufacturing key chemicals, building OSAT capabilities, and scaling up system integration across diverse sectors like automotive, telecommunications, and consumer electronics, India's role is growing stronger each day. With increasing global shifts like China +1, the country is not just catching up but is well on its way to becoming a critical player in the global semiconductor ecosystem. While India may not yet be home to the world's largest semiconductor fabs, its foundational steps in OSAT, chemical manufacturing, and developing a skilled workforce positions it as a hub of innovation, offering both high-quality solutions and cost-effective alternatives. The future looks promising, and India's semiconductor ambitions are set to thrive with global partnerships, a robust ecosystem, and a highly skilled workforce.

Bastion Research Ally - Bespoke Equity Research Services

Helping India's most discerning investors make sharper decisions.

At Bastion Research, we partner with family offices, institutional investors, fund managers, and HNIs to deliver high-conviction, custom equity research on Indian listed companies. Our process is grounded in rigorous business-first analysis, with each project culminating in clear, actionable insights.

Our service is designed for investors who value depth, clarity, and accountability in decision-making

Why Work With Bastion?

Not Pre-Packaged – Every project is built from scratch, mapped to your decision-making process.

Sector-Agnostic, Analyst-Led – We are open to covering almost any business irrespective of its sector or market cap.

Quality with Diversity – Partnering with Bastion Research means gaining a dedicated team of top-tier research analysts, each bringing deep expertise across a wide range of industries and sectors.

Built for Portfolios – We don't publish for headlines. We build conviction for portfolios.

Transparent Process – You see our workings, frameworks, and questions, not just the result.

Clarity Before Action – We keep refining our work until you're confident in the call.

(Click Here)

Request

RESEARCH REPORT & PRODUCT NOTE

Disclaimer & Disclosures

The following Disclosures are being made in compliance with the SEBI Research Analyst Regulations 2014 (herein after referred to as the Regulations)

PARTH L AGRAWAL ("Research Analyst") is registered with SEBI as Research Analyst with Registration No. INH000013712. The research analyst got its SEBI registration on November 29, 2023 and is engaged in research and recommendation Services. The focus of Research Analyst is to provide research and recommendations services to the clients. Analyst aligns its interests with those of the client and seeks to provide the best suited services.

- Research Analyst have no material adverse disciplinary history as on the date of publication of this report.
- Research Analyst has no associates.
- Research Analysts or its employee or its associates may have beneficial ownership of 1% or more in the subject company at the end of the month immediately preceding the date of publication of the Research Report.
- Research Analysts or its employee or its associates does not have any financial interest in the subject company.
- Research Analysts or its employee or its associates have no actual or potential conflicts of interest arising from any connection to or association with any issuer of products/ securities, including any material information or facts that might compromise its objectivity or independence in the carrying on of recommendation services at the time of publication of the research report.
- Research Analysts or its employee or its associates have not received any kind of remuneration or consideration, or compensation from the subject company or from anyone in connection with the research report in the past twelve months.
- The subject company was not a client of Research Analyst during twelve months preceding the date of distribution of the research report.
- Research Analysts or its employee or its associates have not served as an officer, director or employee of the subject company.
- Research Analysts or its employee or its associates have not been engaged in market making activity of the subject company

This report has been prepared by Research Analyst and is solely for information of the recipient only. The report must not be used as a singular basis of any investment decision. The views herein are of a general nature and do not consider the risk appetite or the particular circumstances of an individual investor; readers are requested to take professional advice before investing. Nothing in this document should be construed as investment advice.

Each recipient of this document should make such investigations as they deem necessary to arrive at an independent evaluation of an investment in securities of the companies referred to in this document (including merits and risks) and should consult their own advisors to determine merits and risks of such investment.

The information and opinions contained herein have been compiled or arrived at, based upon information obtained in good faith from sources believed to be reliable. Such information has not been independently verified and no guaranty, representation of warranty, express or implied, is made as to its accuracy, completeness or correctness. All such information and opinions are subject to change without notice. Descriptions of any company or companies or their securities mentioned herein are not intended to be complete. Research Analyst is not obliged to update this report for such changes. Research Analyst has the right to make changes and modifications at any time

Disclaimer & Disclosures

This report is not directed to, or intended for display, downloading, printing, reproducing or for distribution to or use by, any person or entity who is a citizen or resident or located in any locality, state, country or other jurisdiction where such distribution, publication, reproduction, availability or use would be contrary to law or regulation or what would subject Research Analyst or its affiliates to any registration or licensing requirement within such jurisdiction. If this report is inadvertently sent or has reached any person in such country, especially, United States of America, the same should be ignored and brought to the attention of the sender. This document may not be reproduced, distributed or published in whole or in part, directly or indirectly, for any purposes or in any manner.

It should not be considered to be taken as an offer to sell or a solicitation to buy any security. This document is not, and should not, be construed as an offer or solicitation of an offer, to buy or sell any securities or other financial instruments. This report should not be construed as an invitation or solicitation to do business with Research Analyst.

Research Analyst do not take any responsibility, financial or otherwise, of the losses or the damages sustained due to the investments made or any action taken on basis of this report.

Investment in securities market are subject to market risks. Read all the related documents carefully before investing. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors.

Statutory Disclosures:

PARTH L AGRAWAL

Registered Address: UG – 64, VIP Plazza, Near Shyam Baba Mandir, VIP Road, Vesu, Surat

Registration No.: INH000013712

Contact No.: 91-8780507966

Email: connect@bastionresearch.in

Statutory Disclaimers:

"Investment in securities market is subject to market risks. Read all the related documents carefully before investing. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors